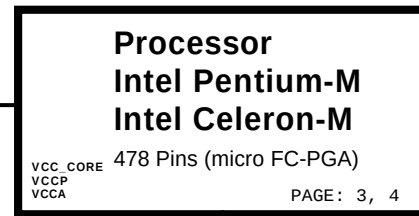


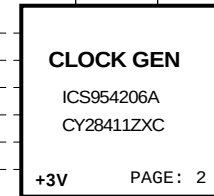
PENTIUM-M / ALVISO / ICH6-M

LAYER 1 : TOP
LAYER 2 : GND
LAYER 3 : IN1
LAYER 4 : IN2
LAYER 5 : VCC
LAYER 6 : BOT

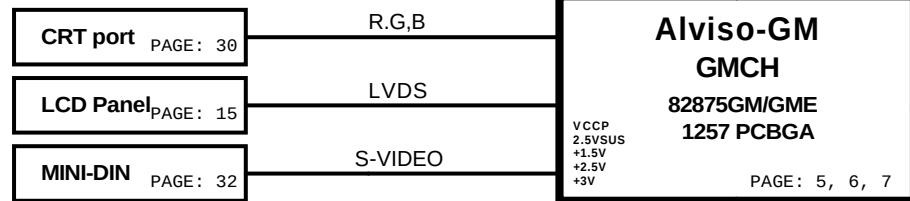
**CPU THERMAL
SENSOR**
MAX6657 / GMT-781
+3V PAGE: 3



```
HCLK_CPU
HCLK_CPU#
HCLK_MCH
HCLK_MCH#
SRC_MCH
SRC_MCH#
SRC_ICH
SRC_ICH#
DREFSSCLK
DREFSSCLK#
DOT96
DOT96#
```

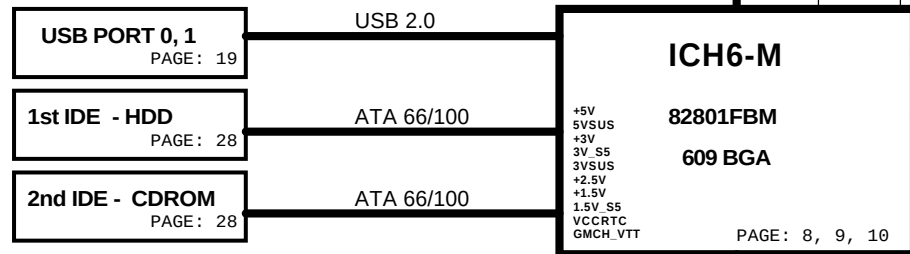


DISCHARGE PAGE: 35

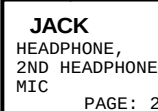
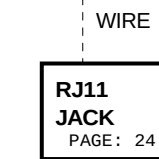
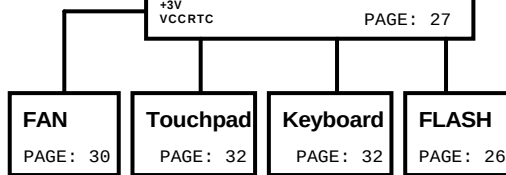
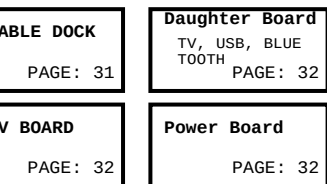
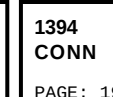
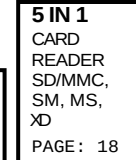
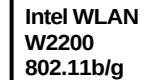
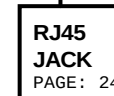
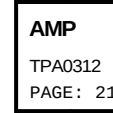
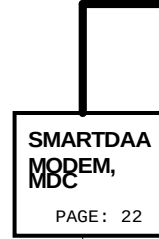
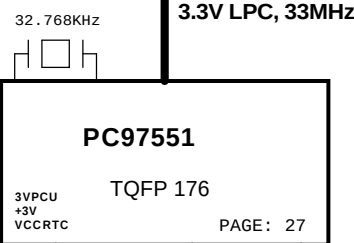
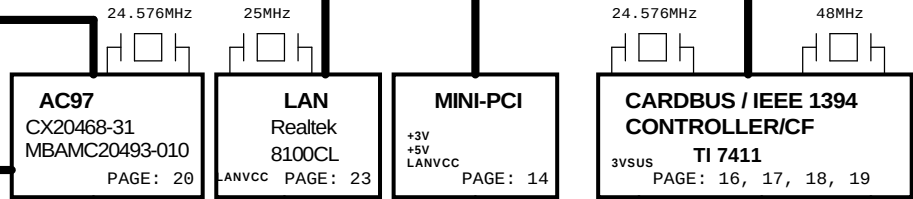


DDR-SODIMM1
2.5VSUS
SMDDR_VTERM
DDR-SODIMM2
PAGE: 13

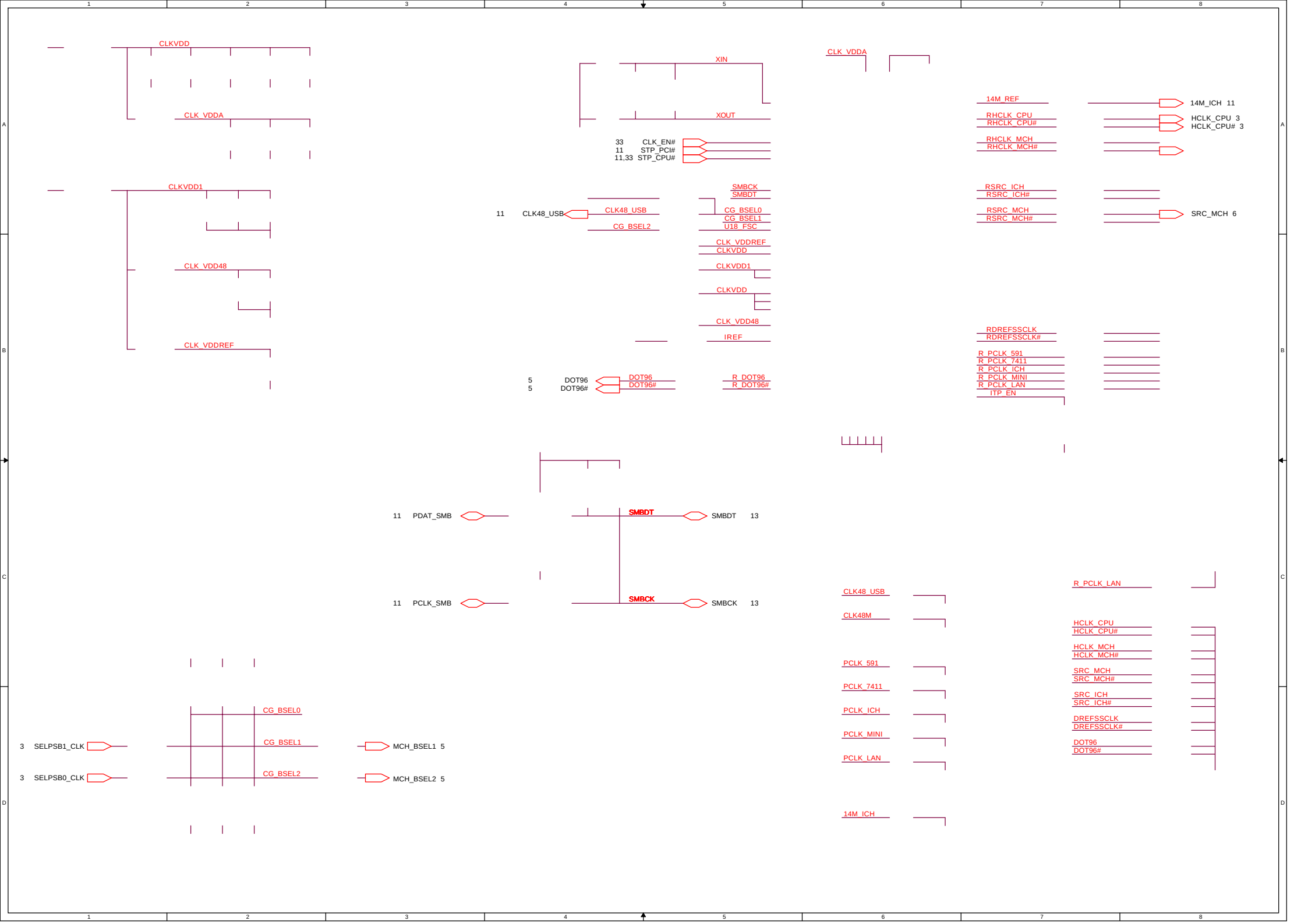
32.768KHz

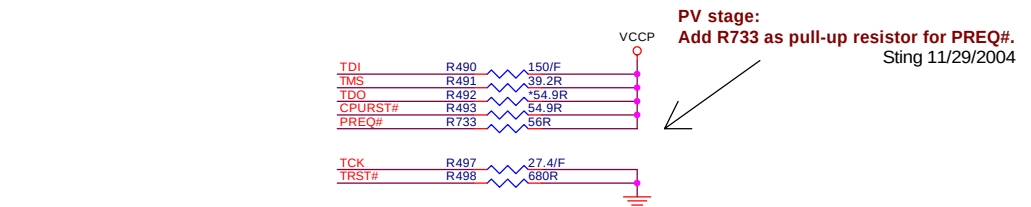
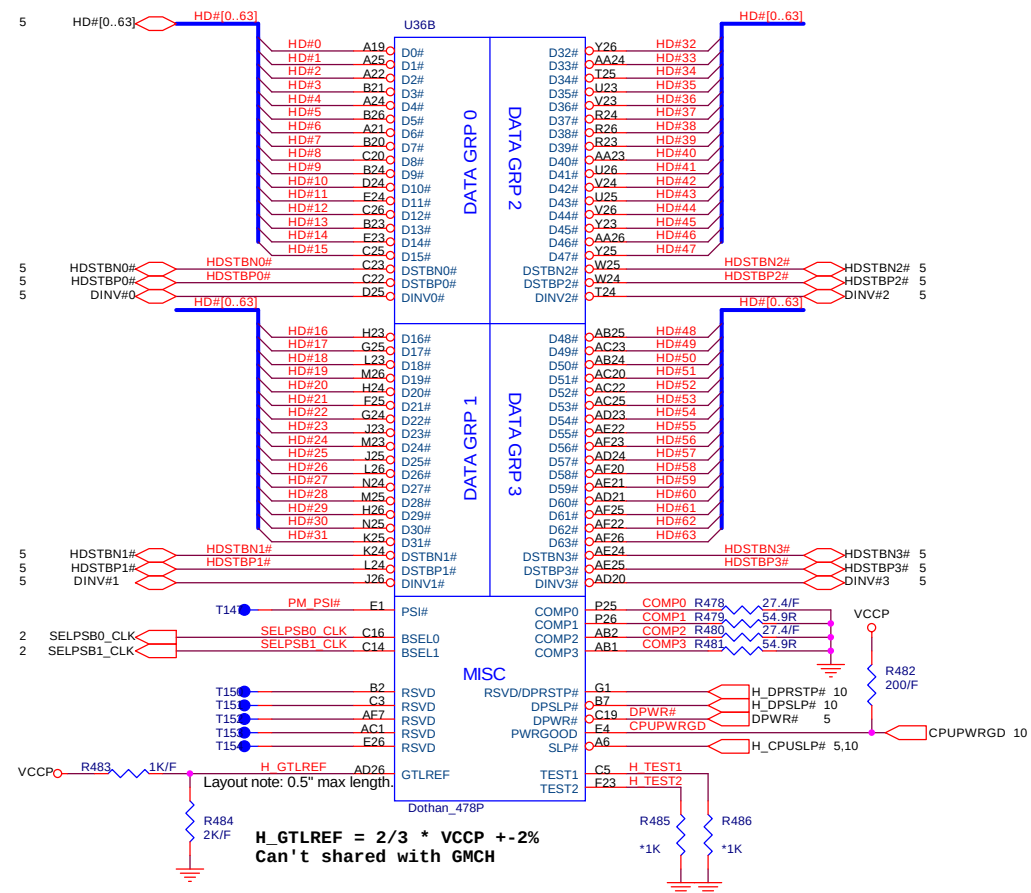
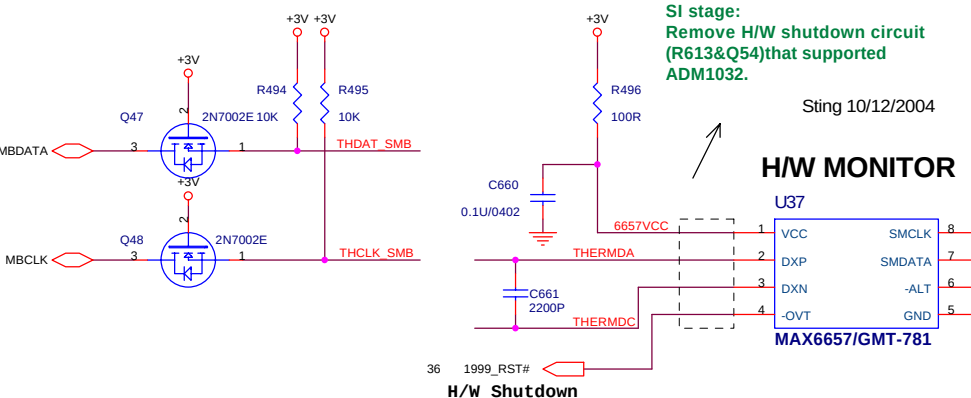


AC97

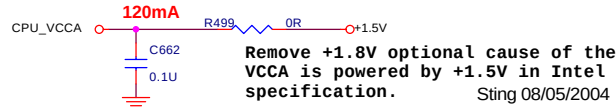
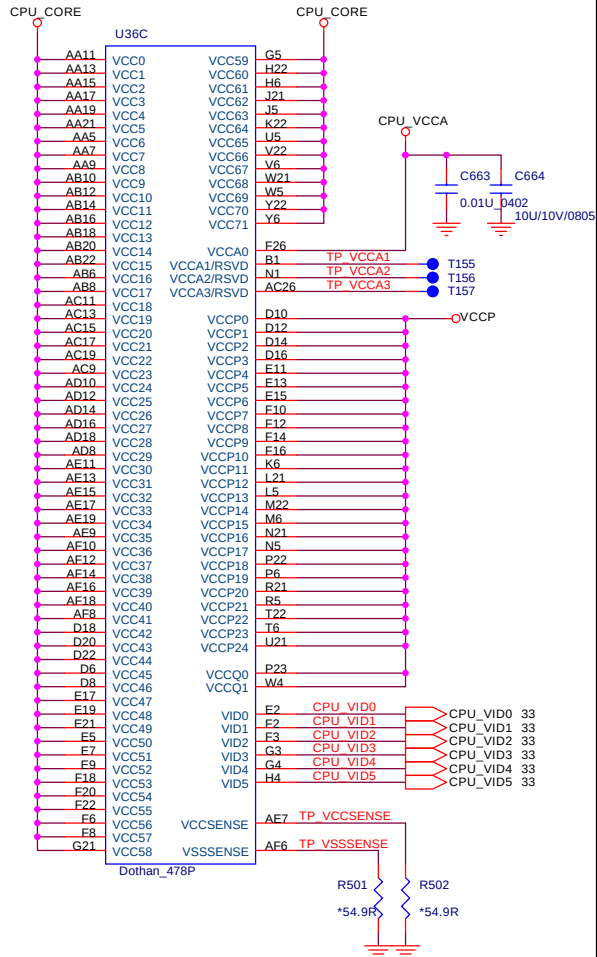


DEVICE	IDSEL #	REQ/GNT #	PCI_INT
GBIT ETHERNET AD16		2	A
MINIPCI SLOT AD22		1	C,D
CardBus/1394 AD25		0	E,F,G

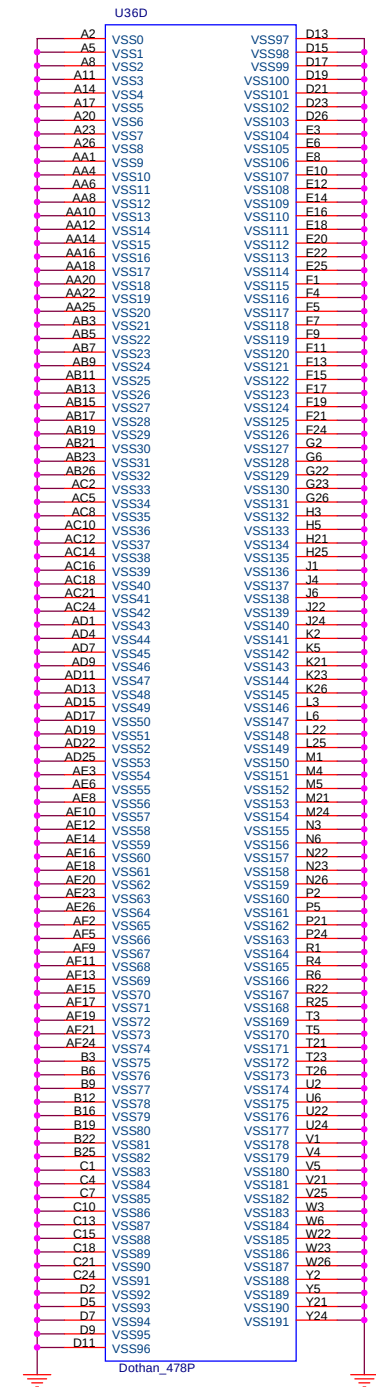
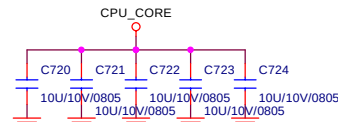
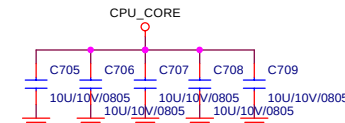
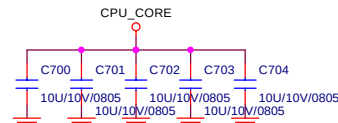
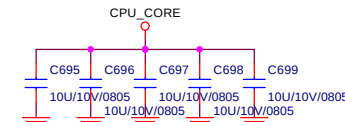
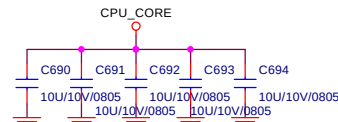
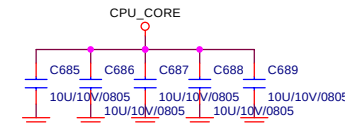
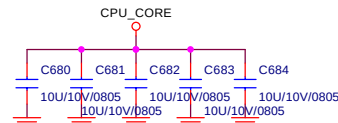
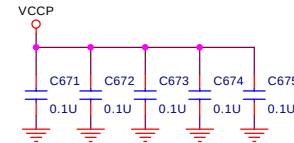
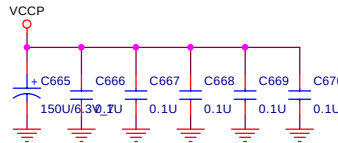




ICH_THRM#
To SB --> System throttling



SI stage:
Use X7R type to replace Y5V type
for Decoupling/Bypass capacitor.
Sting 10/12/2004

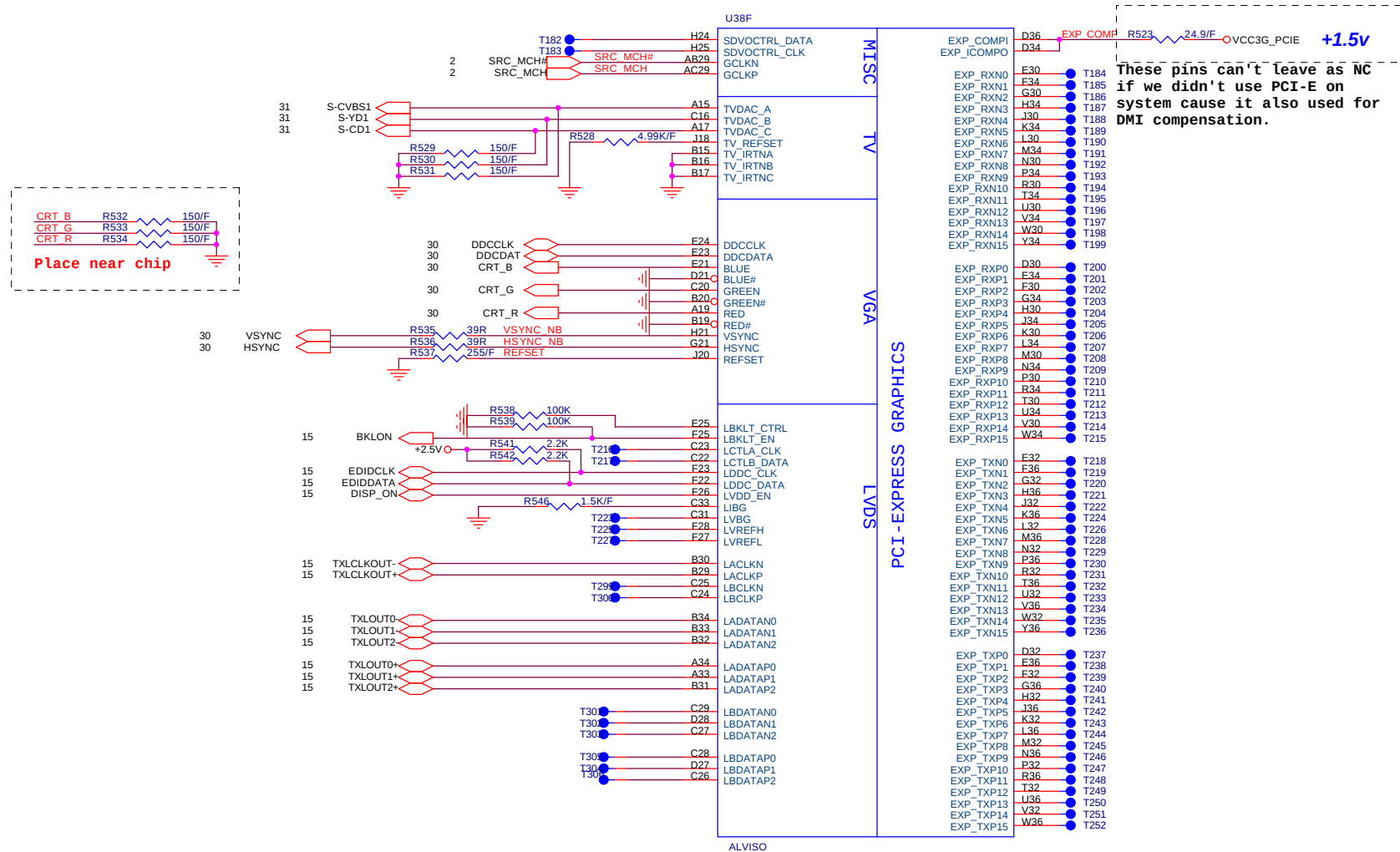


PROJECT : CT3
Quanta Computer Inc.

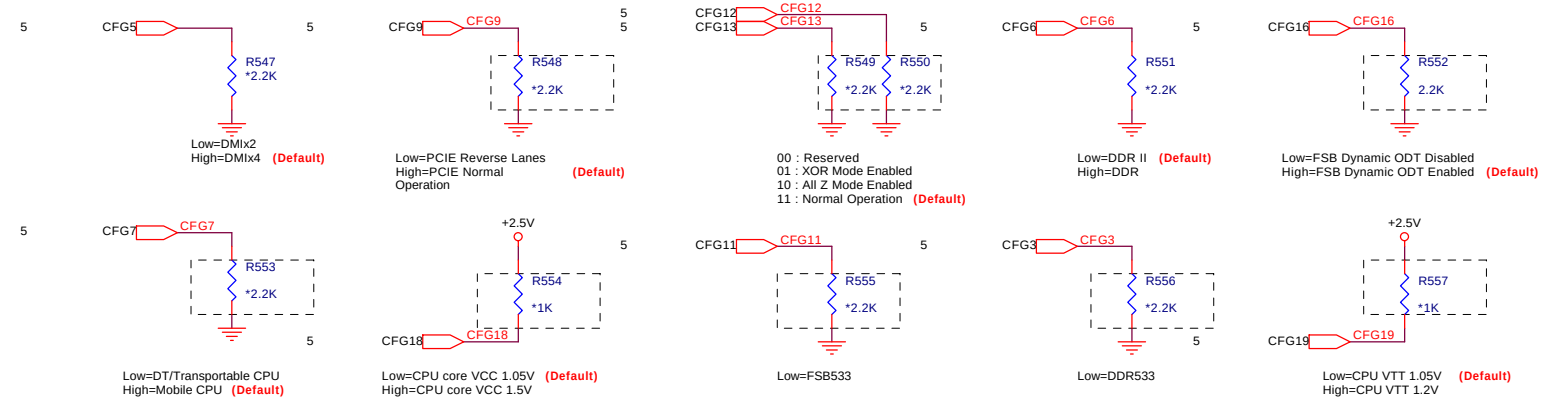
Size	Document Number	Rev
Custom	CPU POWER / GND	2A
Date:	Friday, March 04, 2005	Sheet 4 of 42

CPU Bypass
capacitor

CPU de-coupling
capacitor

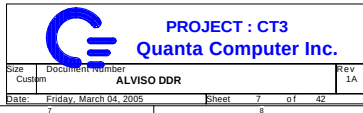


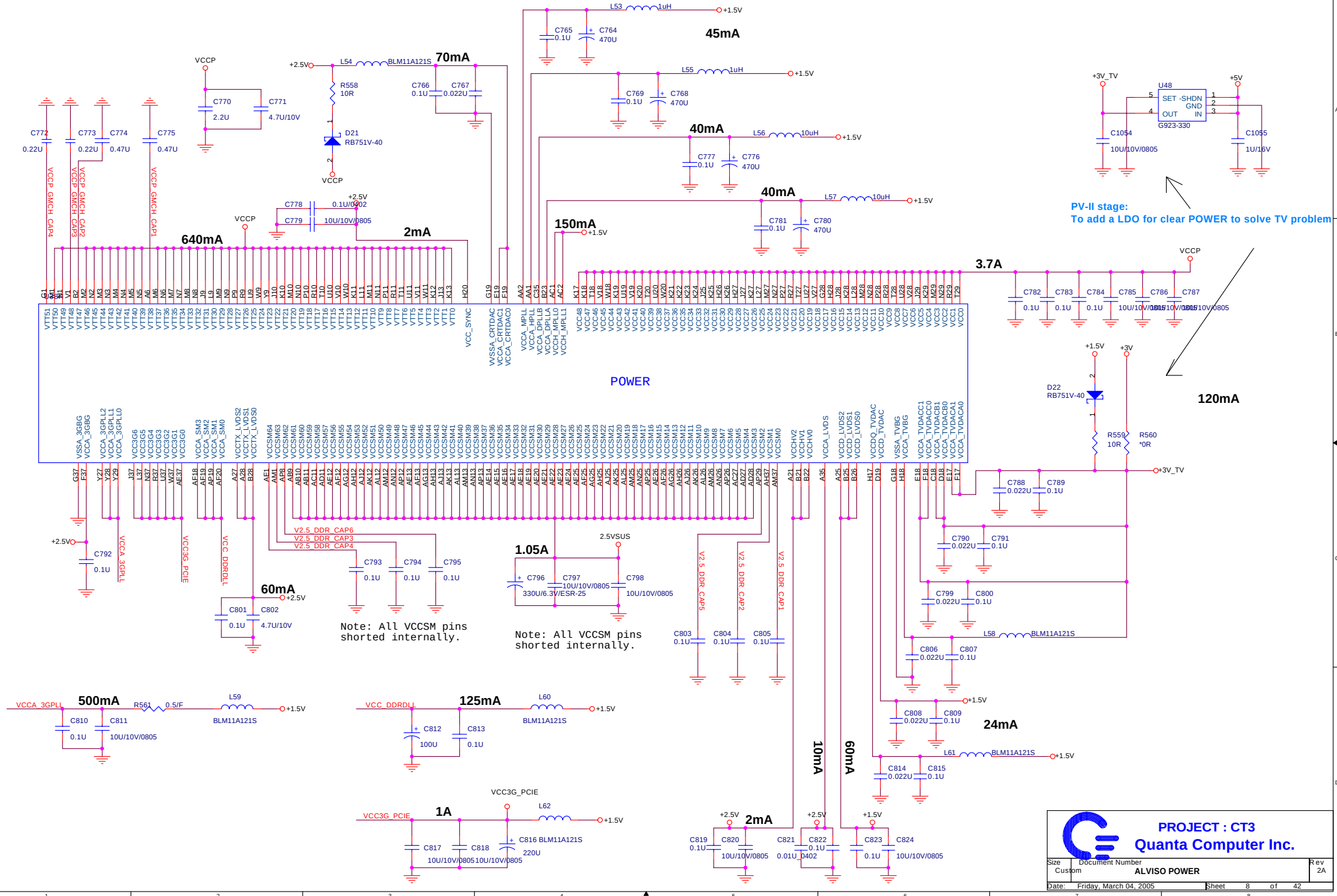
Strapping

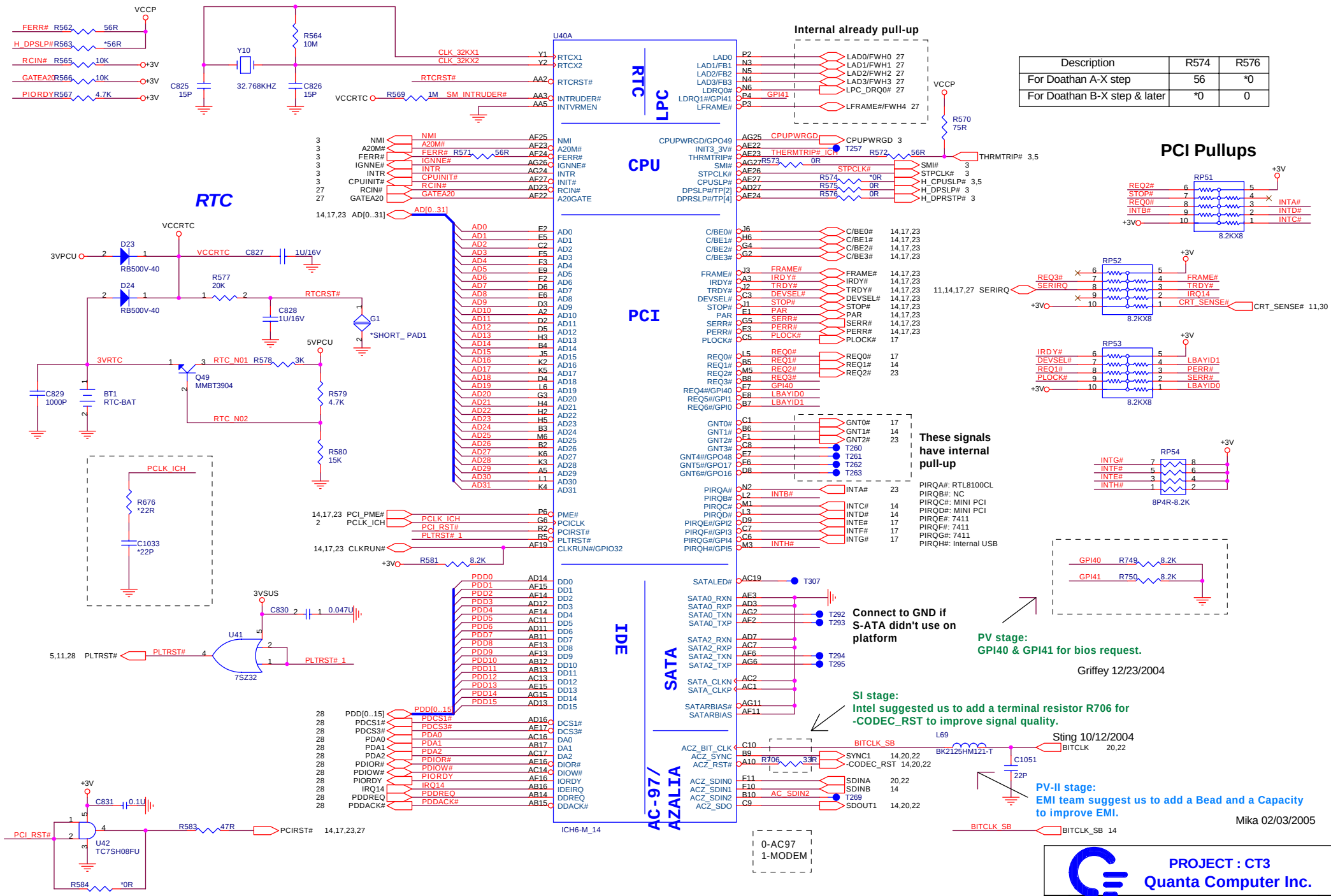


CFG[2:0]
001=533MT/S FSB
101=400MT/S FSB

CFG[3:17] have internal pullup.
CFG[18:19] have internal pulldown.

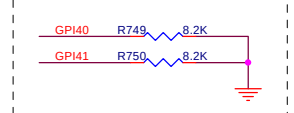
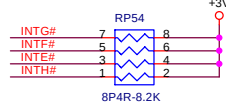
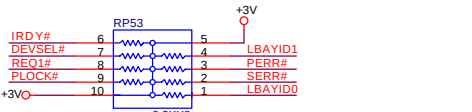
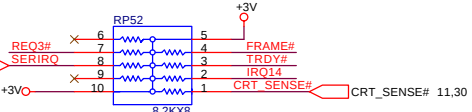
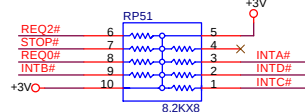






Description	R574	R576
For Doathan A-X step	56	*0
For Doathan B-X step & later	*0	0

PCI Pullups



These signals have internal pull-up

Connect to GND if S-ATA didn't use on platform

SI stage: Intel suggested us to add a terminal resistor R706 for -CODEC_RST to improve signal quality.

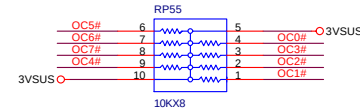
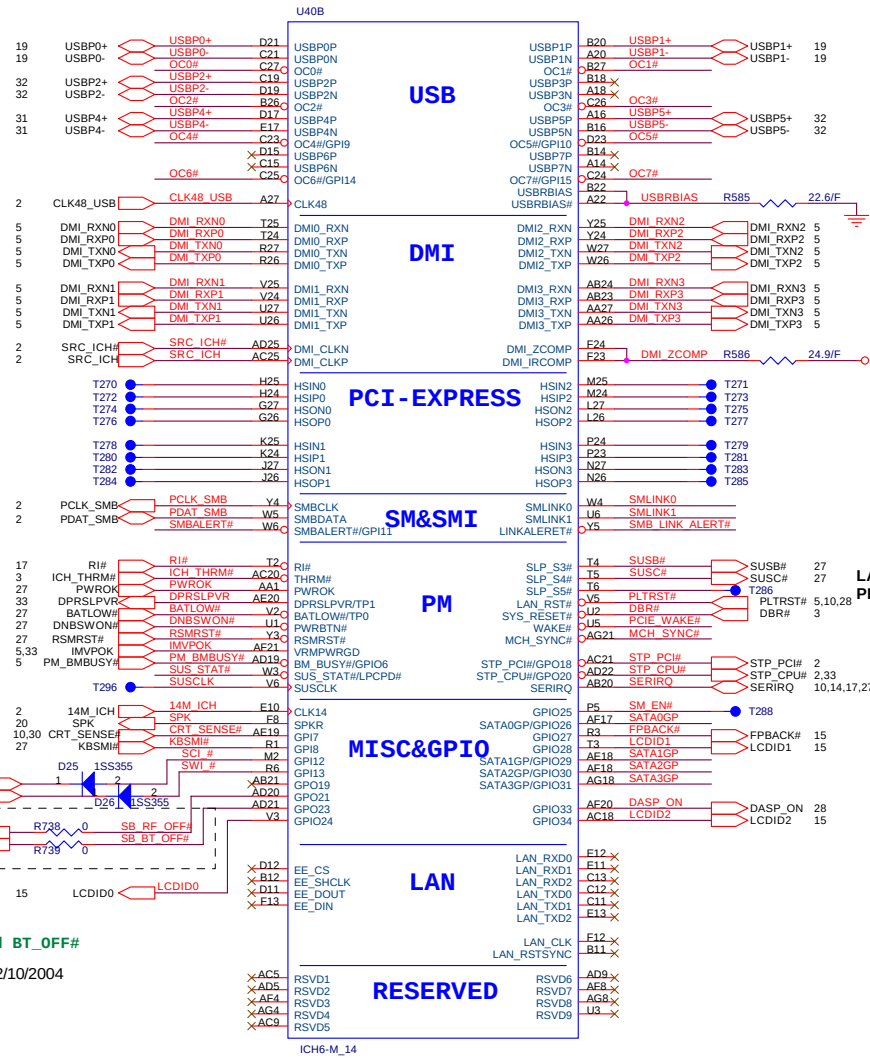
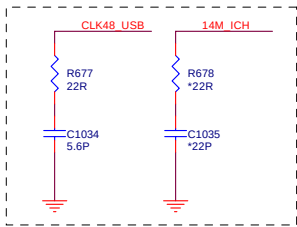
PV stage: GPI40 & GPI41 for bios request.

Griffey 12/23/2004

Sting 10/12/2004

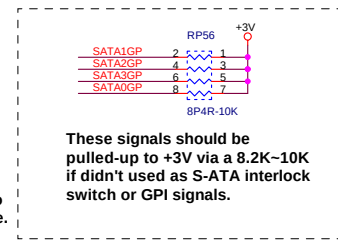
PV-II stage: EMI team suggest us to add a Bead and a Capacity to improve EMI.

Mika 02/03/2005

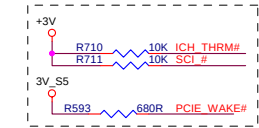
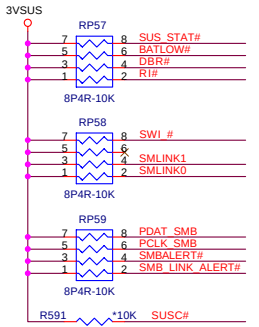
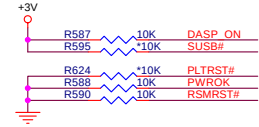


Place within 500mils of ICH-6

Place within 500mils of ICH-6

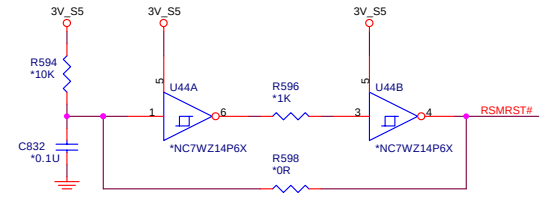


LAN_RST# should be connected to PLTRST# if internal LAN didn't use.



PV stage:
1. Add RF_OFF# and BT_OFF# circuit.

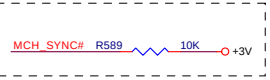
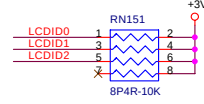
Griffey 12/10/2004



1. Change the power plane of PCIE_WAKE# from 3VSUS to 3V_S5 to solve system can't turn off issue.
2. Change the power plane of ICH_THRM# and SCI_# from 3VSUS to +3V to solve leakage issue.

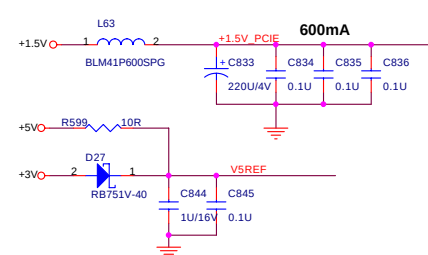
Sting 10/06/2004

Enable Cable ID



SI stage:
R589 should be populated, because MCH_SYNC# is internally ANDed with PWROK. System will not booting without this pulled-up resistor.

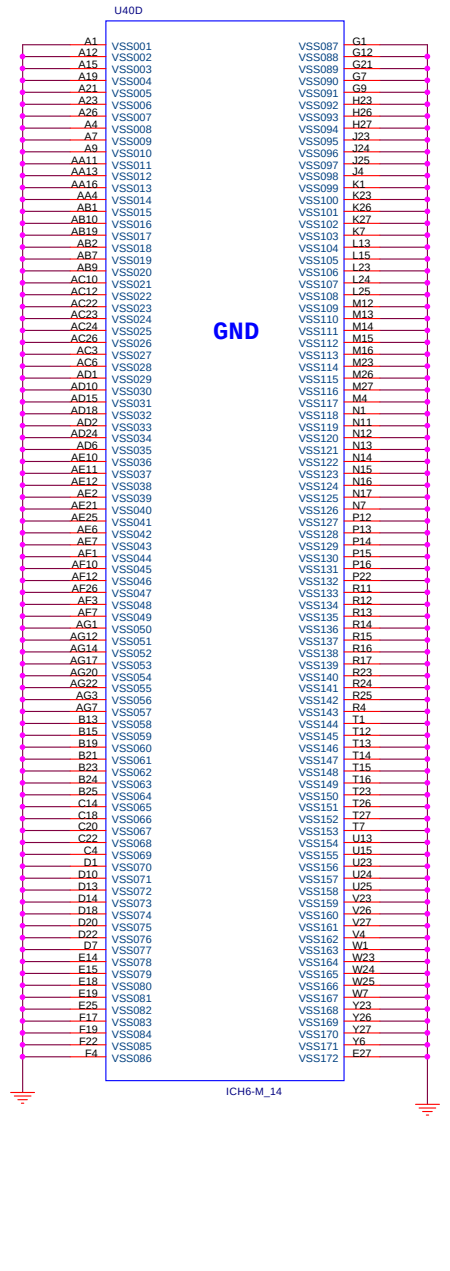
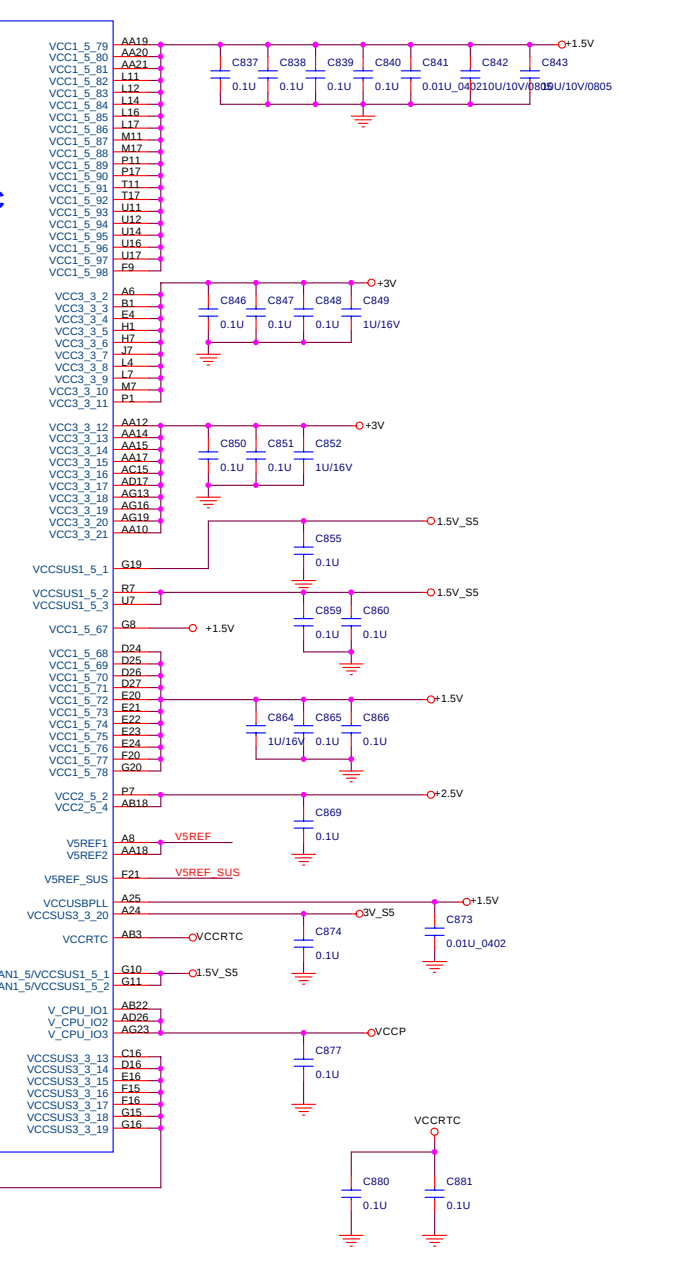
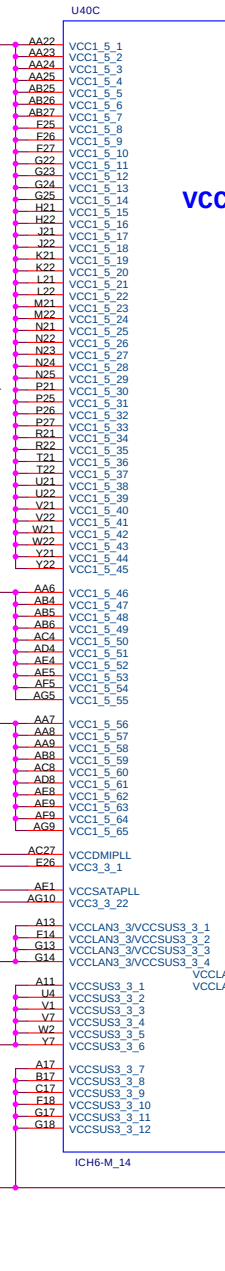
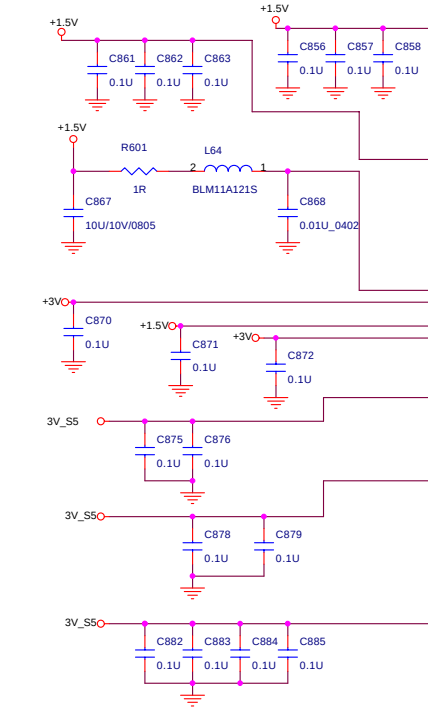
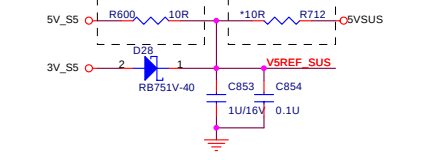
Sting 09/24/2004

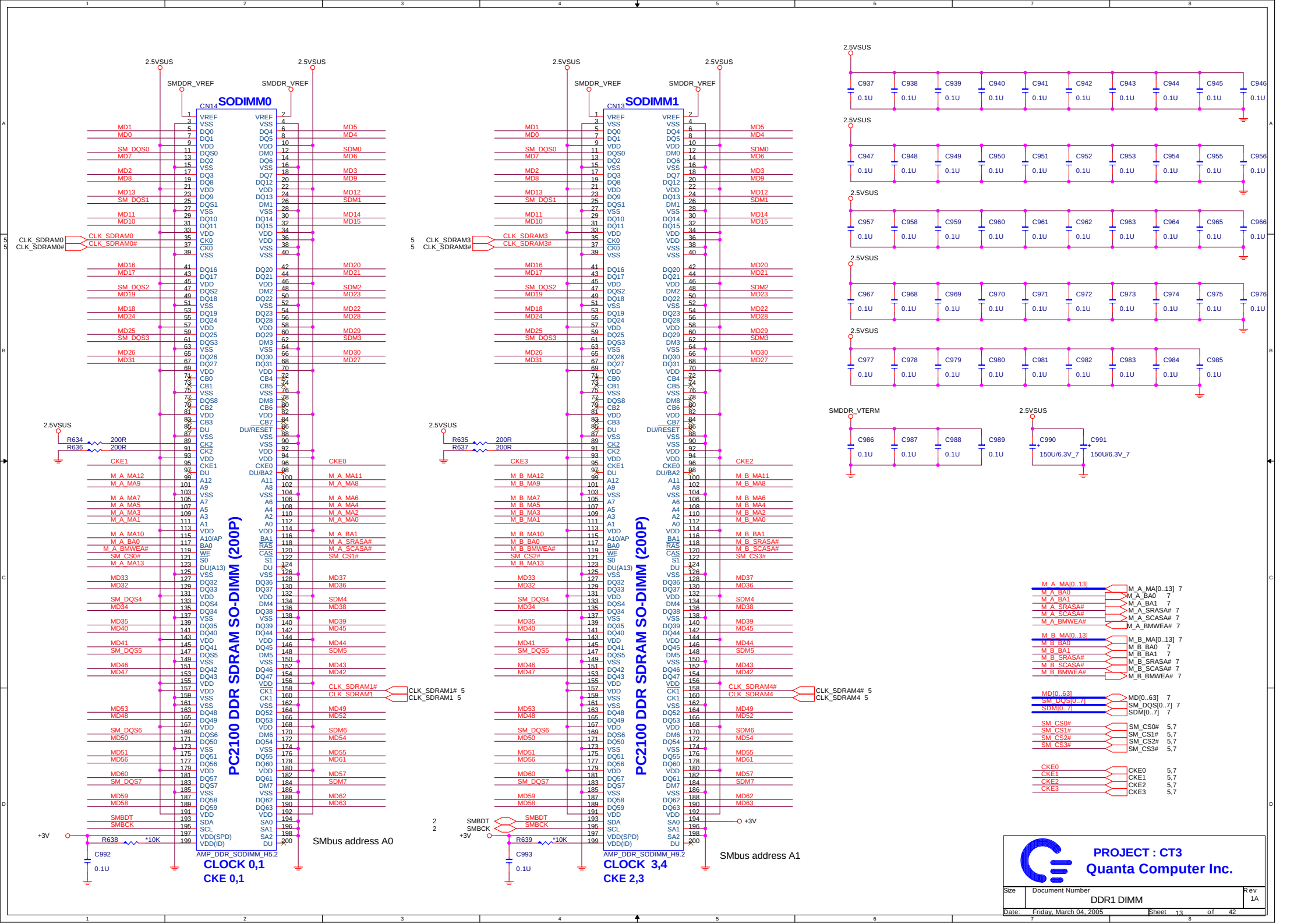


PV stage:
The root cause that is 3V_S5 leakage current to 5VSUS by R712 on S4/S5 stage.

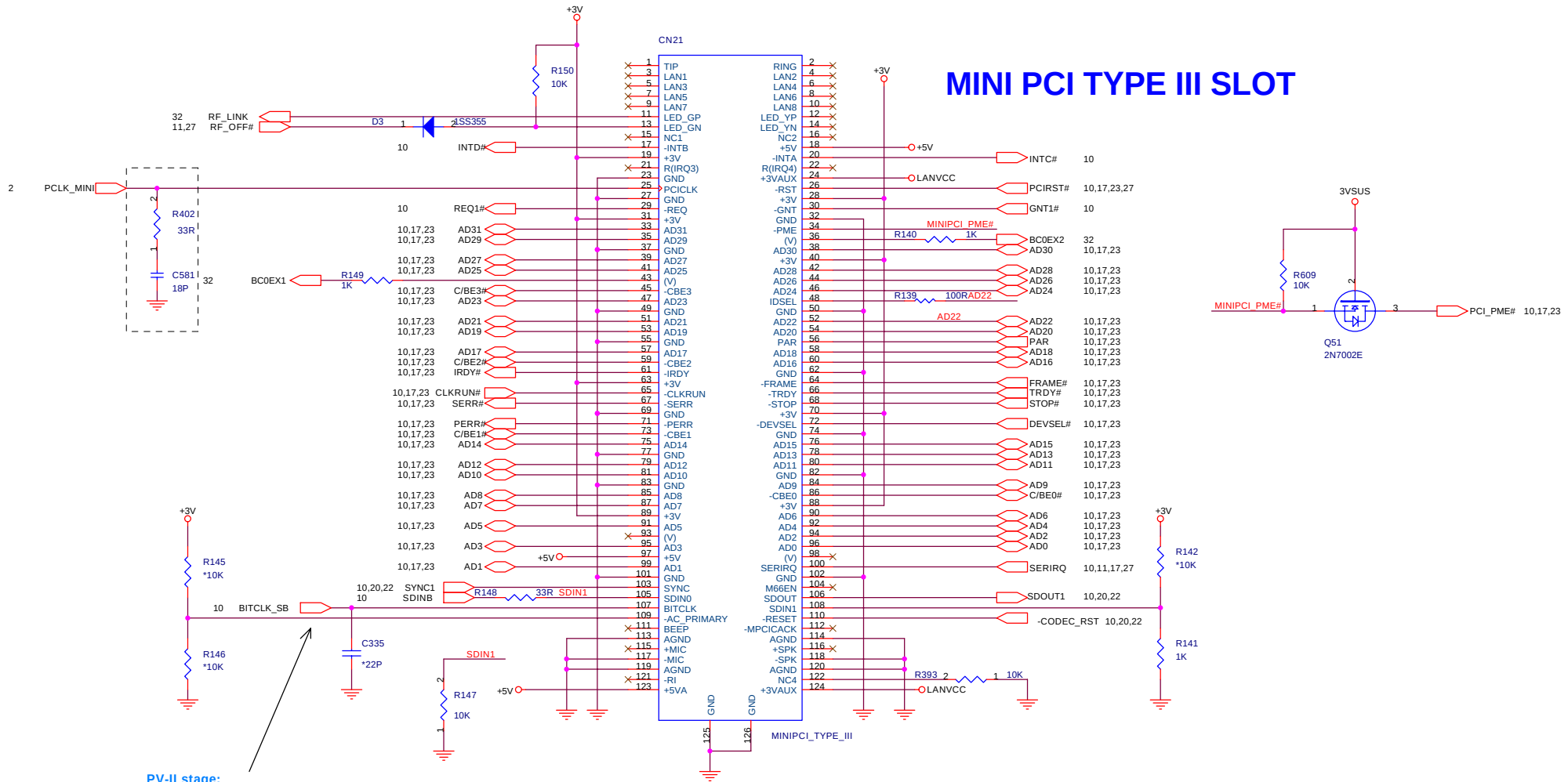
SI stage:
We can also use 5VSUS to instead of 5V_S5 to save cost if 5V_S5 isn't implementation on system.

Griffey 12/13/2004
Sting 10/12/2004



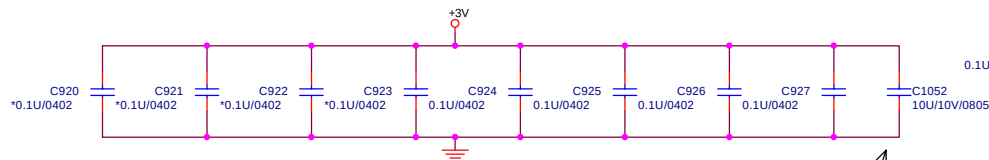
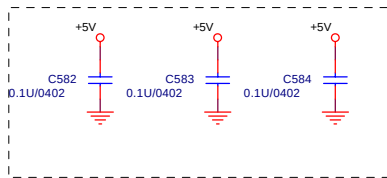


MINI PCI TYPE III SLOT



PV-II stage:
Change net name from BITCLK to BITCLK_SB.

Mika 02/03/2005

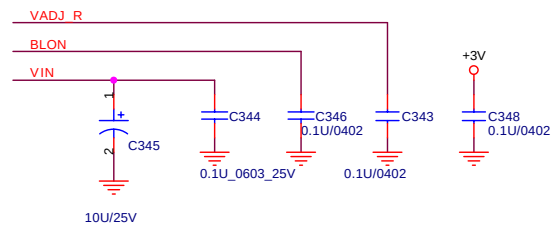
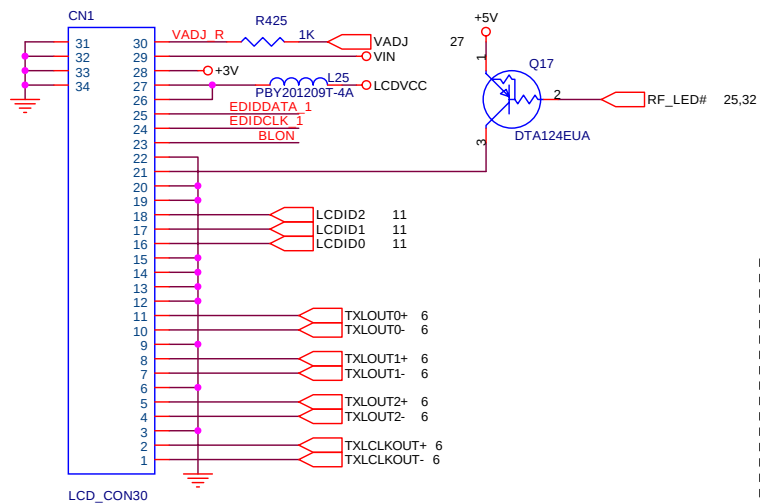


PV-II stage:
Add a 10U Capacity for better power.

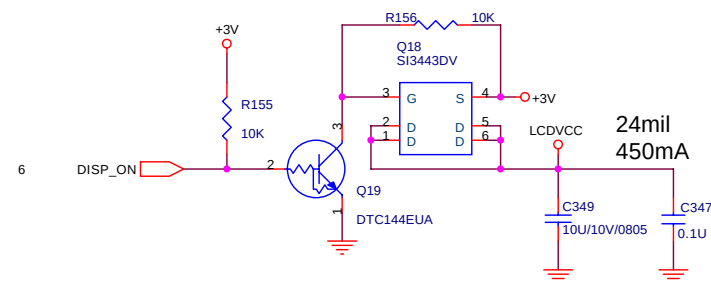
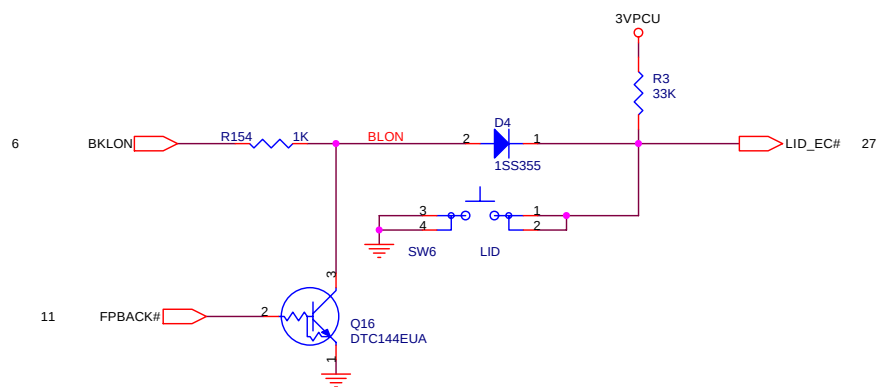
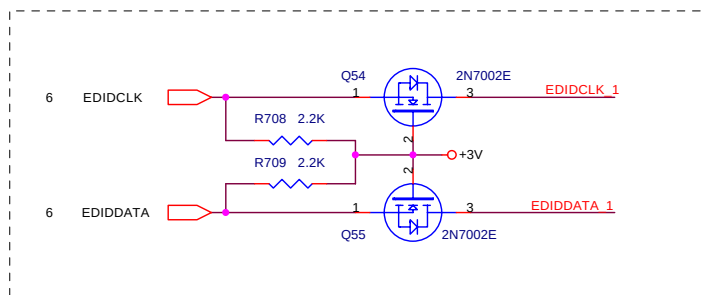
Mika 02/03/2005

PROJECT : CT3
Quanta Computer Inc.

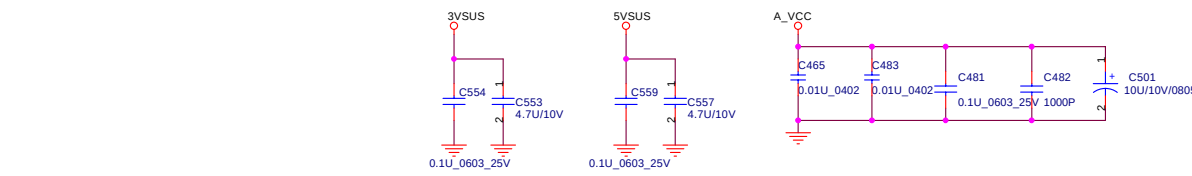
Size	Document Number	Rev
Custom	MINIPCI_TYPE_III	2A
Date:	Friday, March 04, 2005	Sheet 14 of 42



SI stage:
Add a level-shift circuit for EDID interface.
Sting 10/04/2004

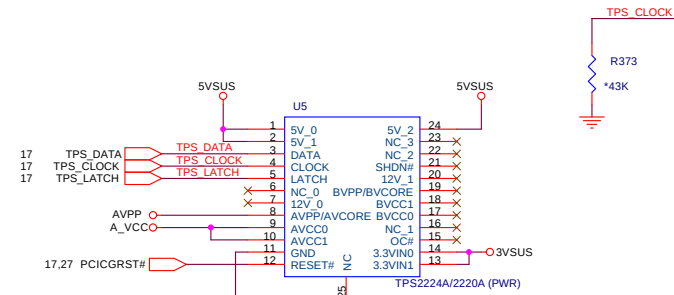


CardBus Connector



CARDBUS POWER SWITCH

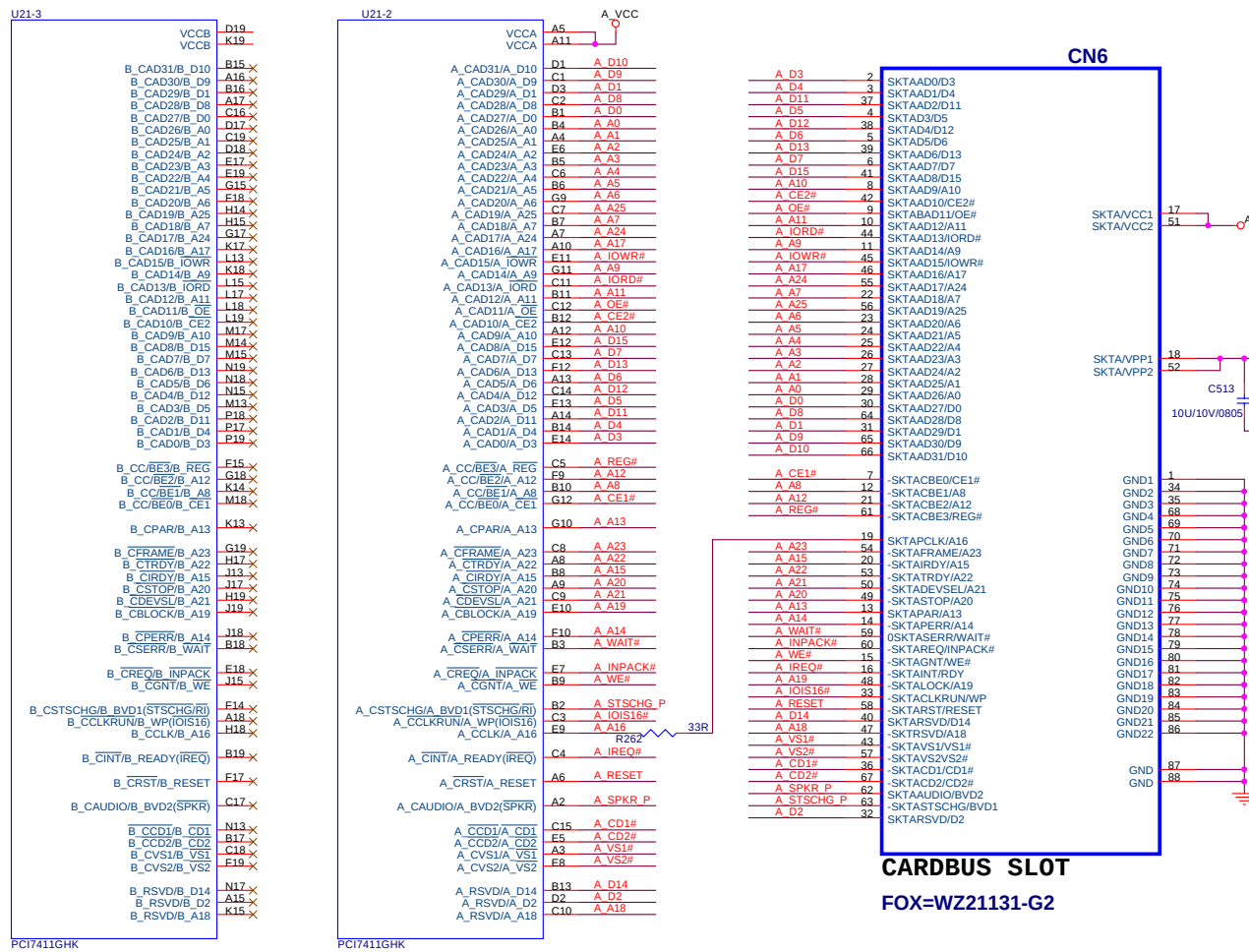
For PCI7411



For PCI1510

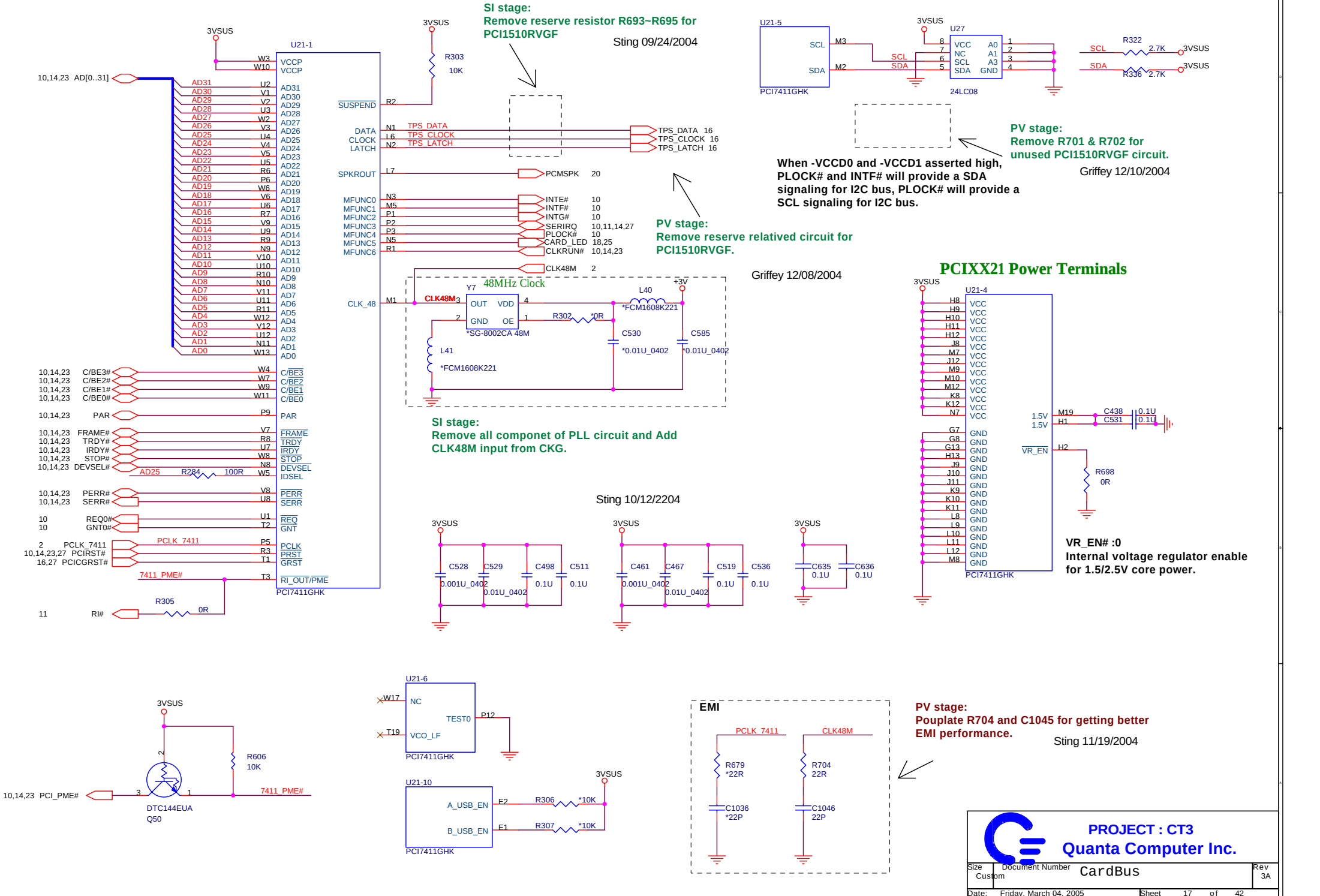
PV stage:
Remove unused PCI1510RVGF circuit.

Griffey 12/08/2004



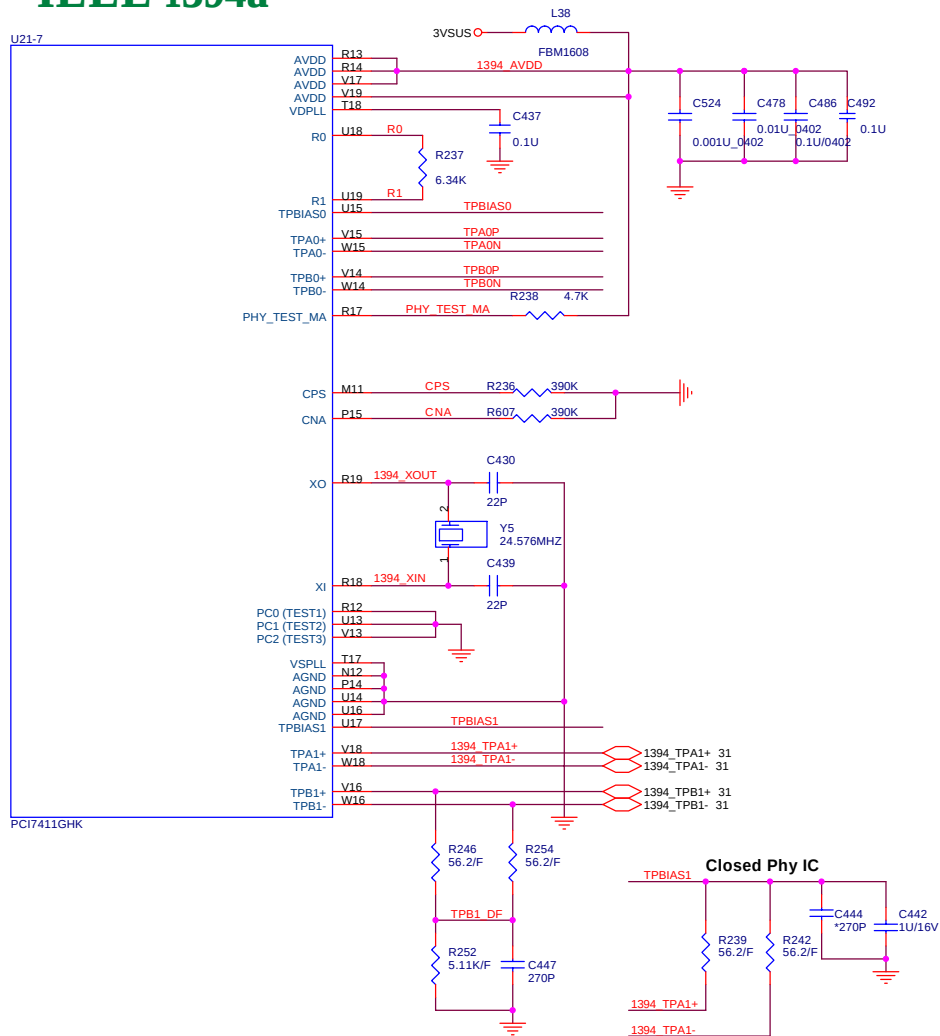
CARDBUS SLOT
FOX=WZ21131-G2

CardBus

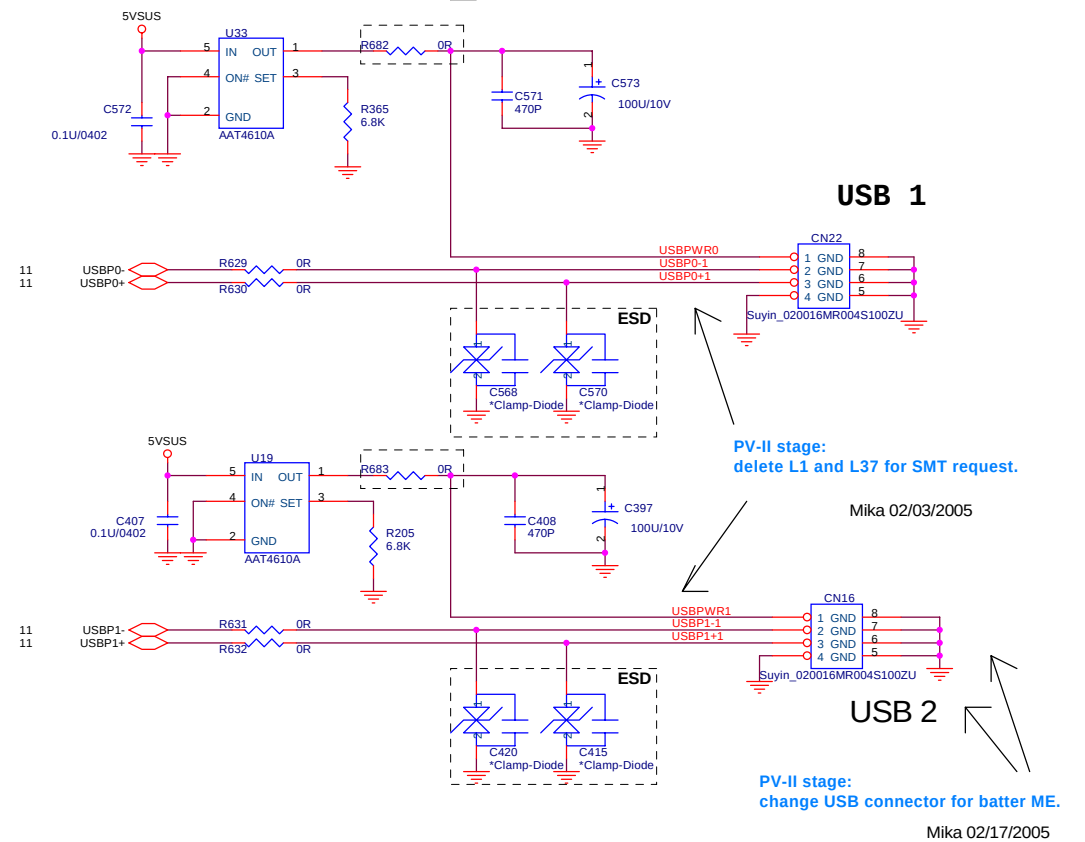
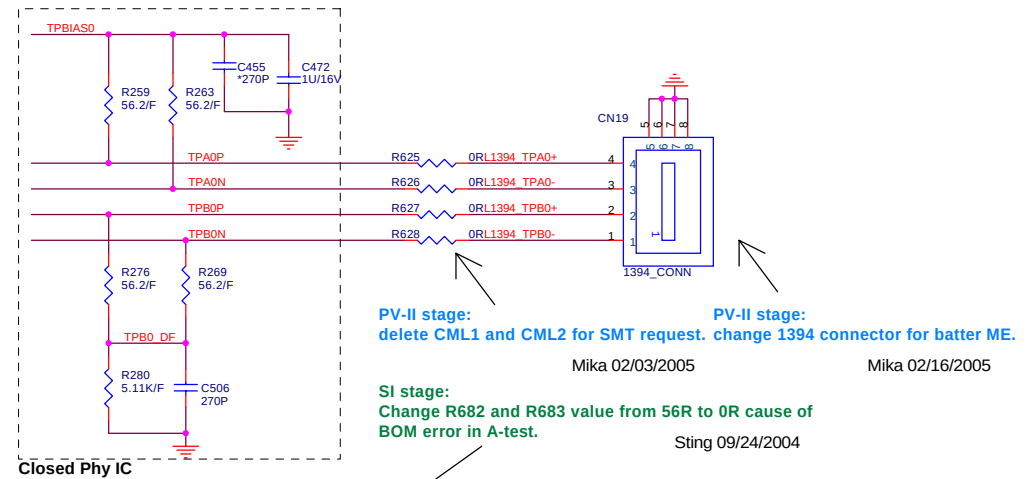




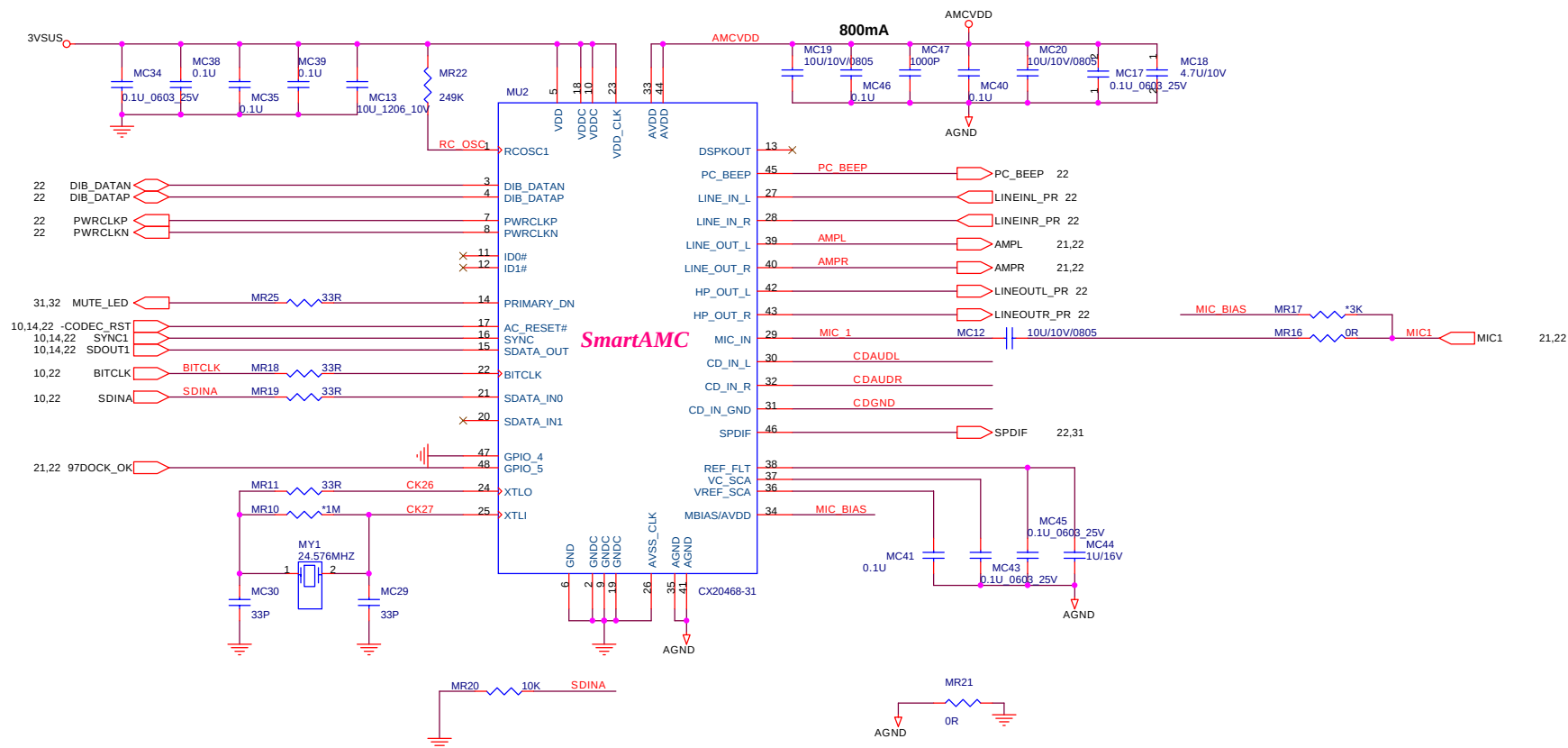
IEEE 1394a



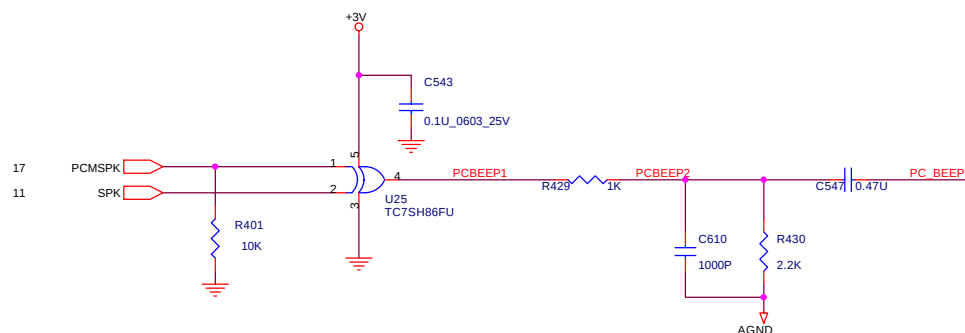
IEEE 1394 CONNECTOR



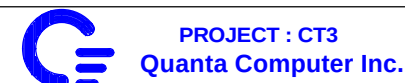
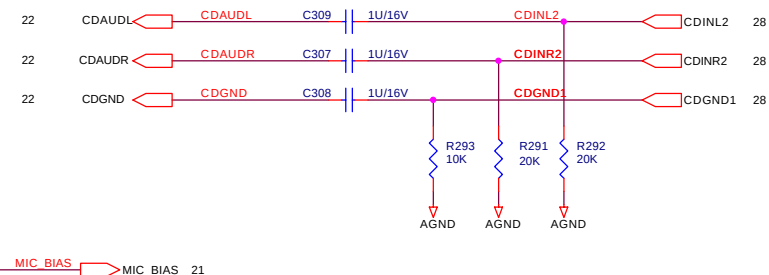
The AMC20493-001 modem is used for mother board family MBAMC20493-010.



PC SPEAKER

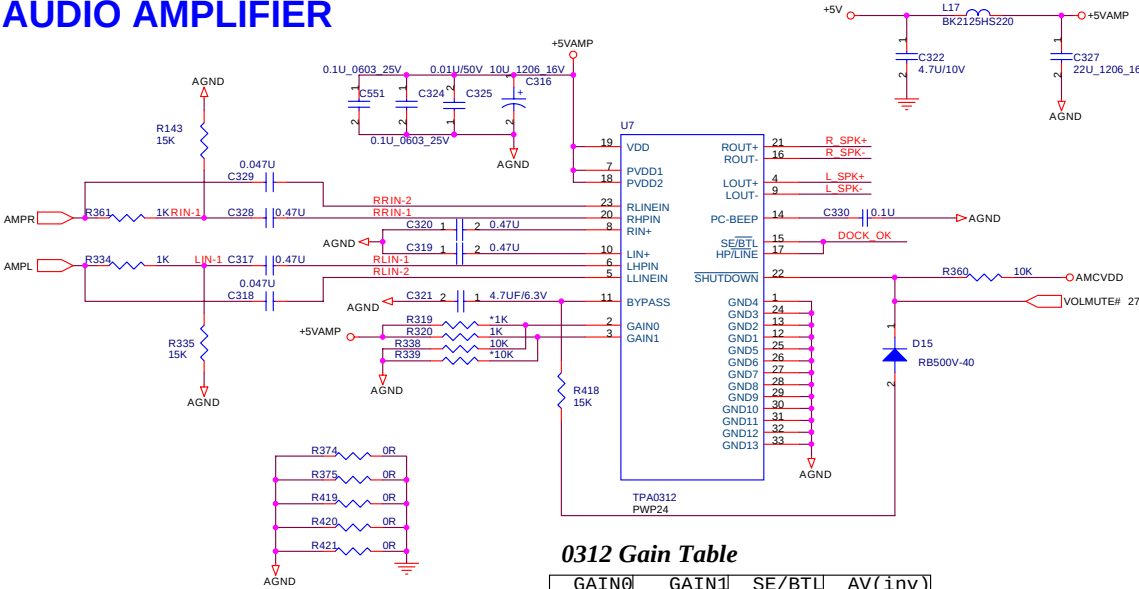


FROM CD-ROM



Size	Document Number	Rev
	AMC97 CODEC	1A
Date:	Friday, March 04, 2005	Sheet 20 of 42

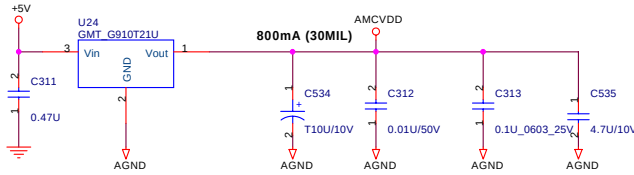
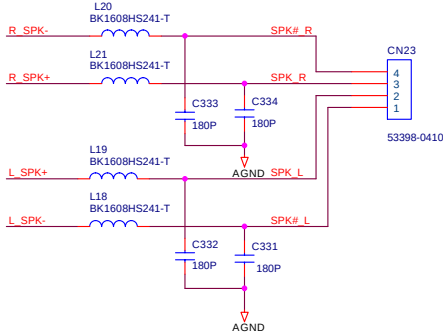
AUDIO AMPLIFIER



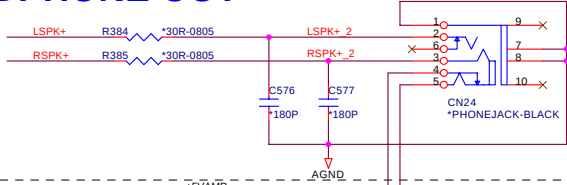
0312 Gain Table

GAIN0	GAIN1	SE/BTL	AV(inv)
0	0	0	6 dB
0	1	0	10 dB
1	0	0	15.6 dB
1	1	0	21.6 dB
X	X	1	4.1 dB

SPEAKER OUT

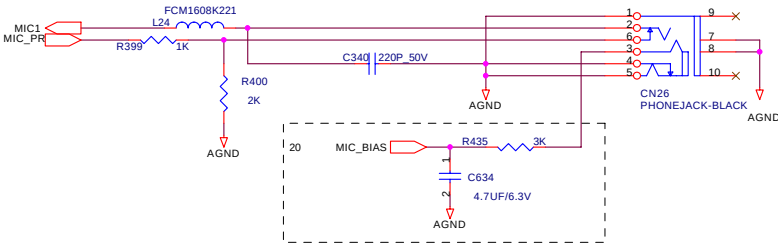


2ND HEADPHONE OUT



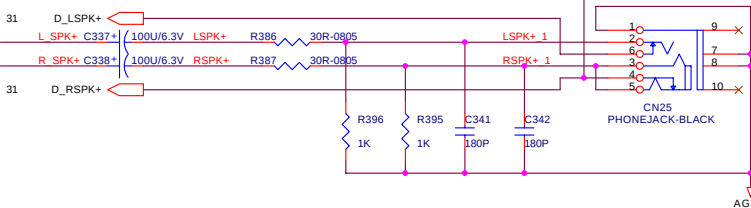
PAVILION

MICROPHONE



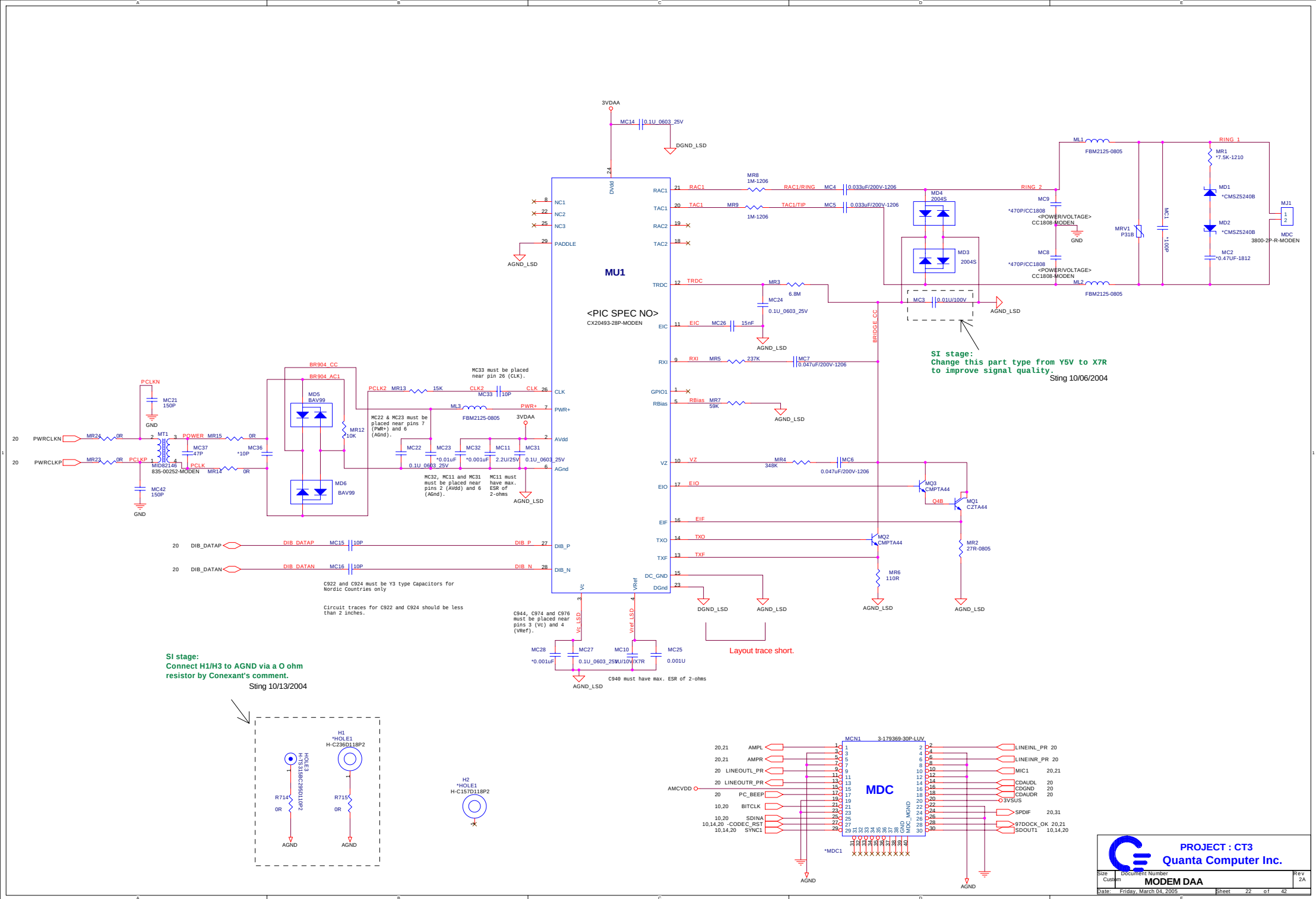
PAVILION

HEADPHONE OUT

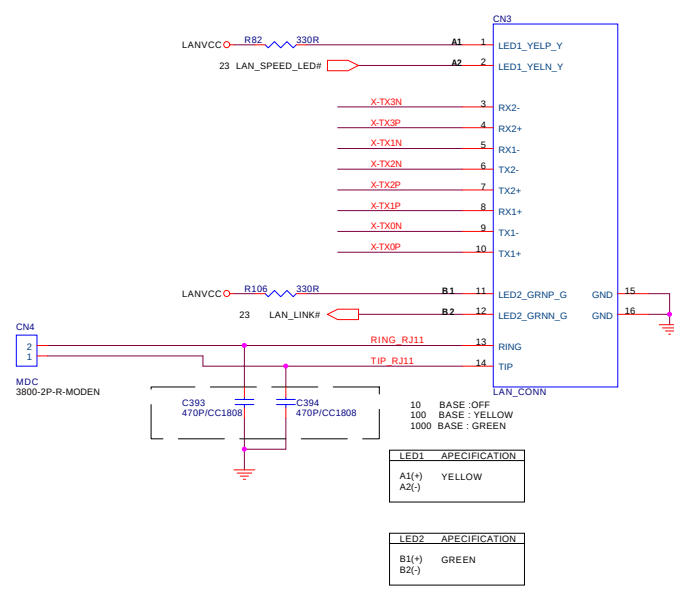
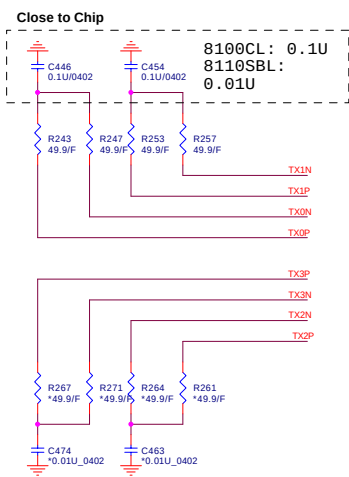
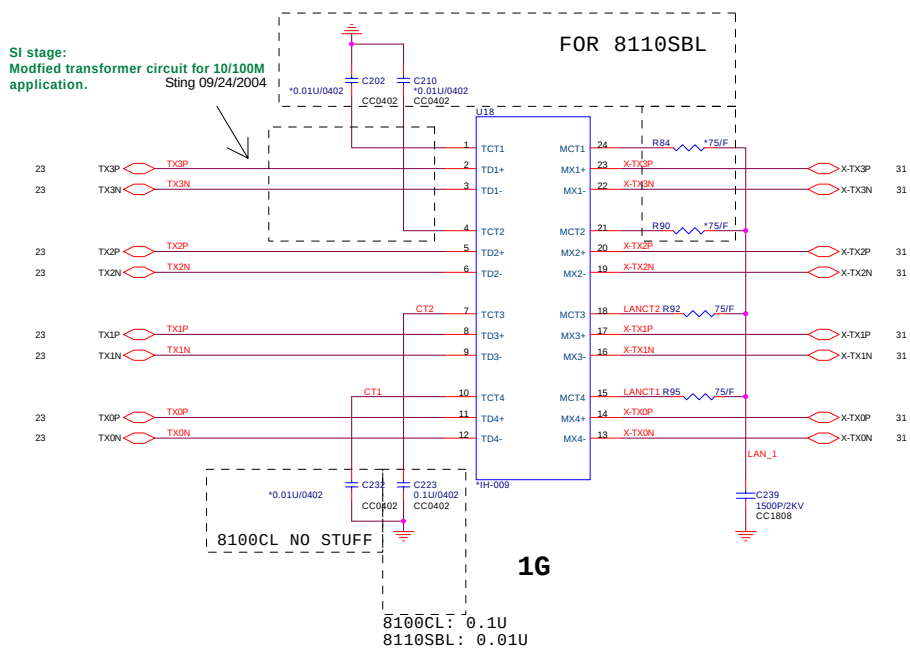


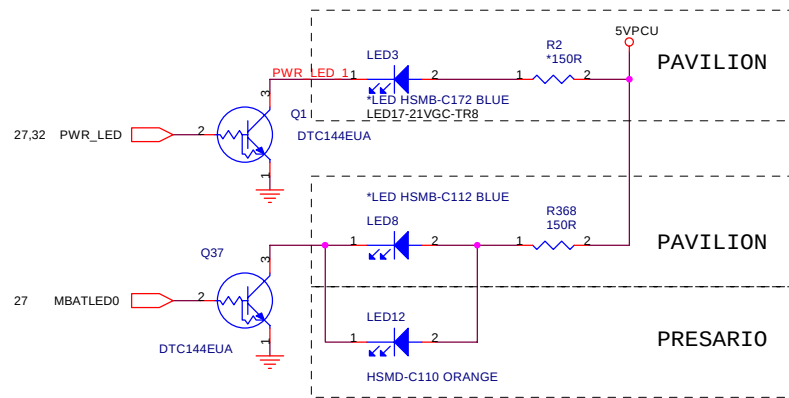
PAVILION

PRESARIO

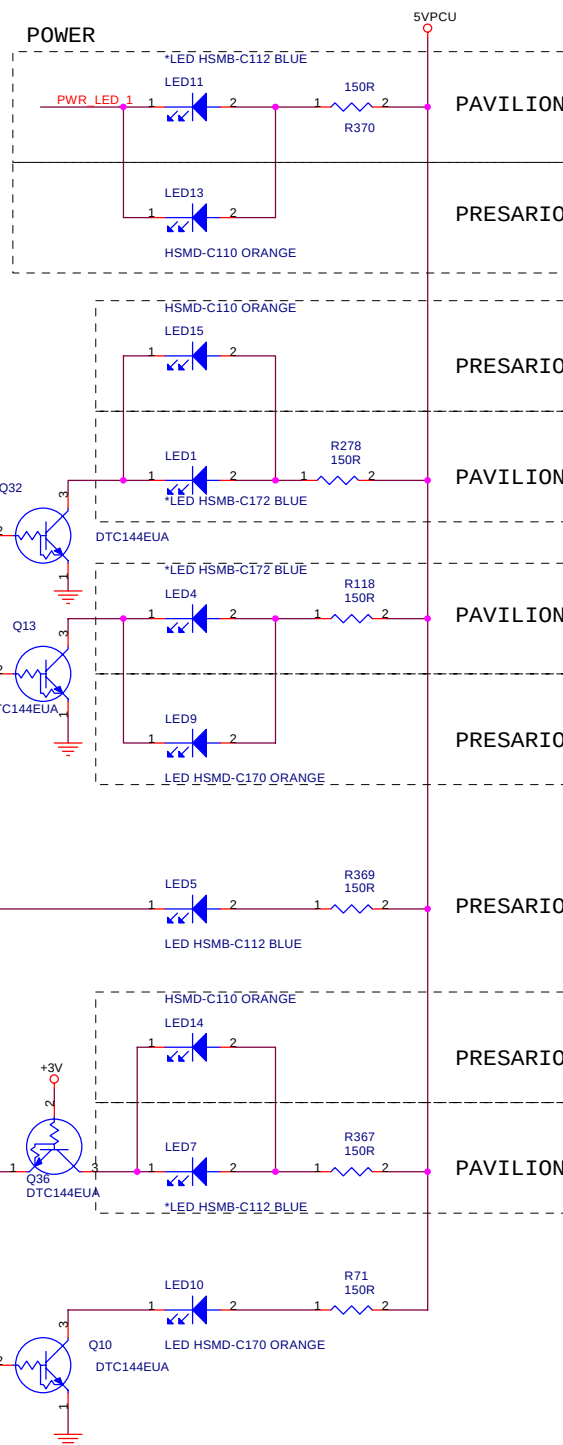
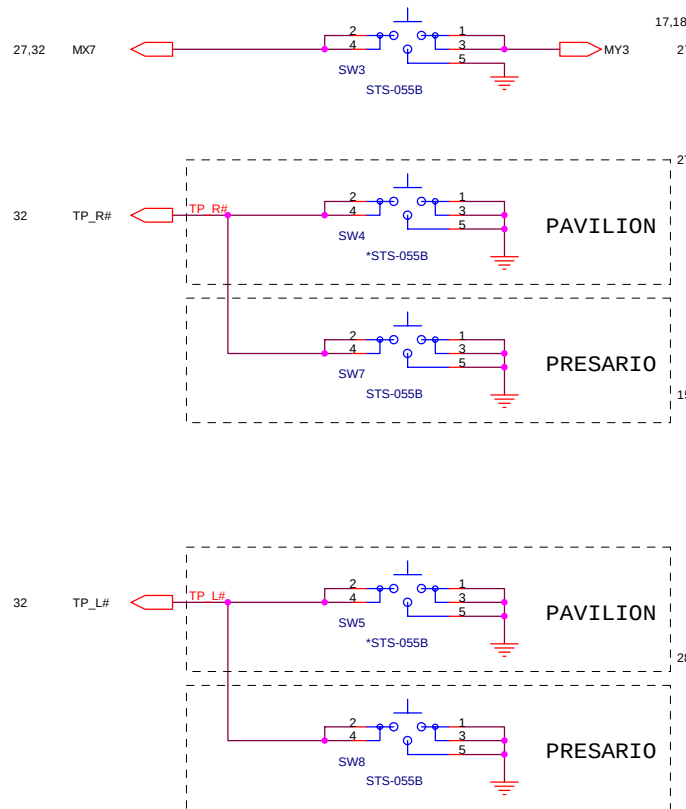


SI stage:
Modified transformer circuit for 10/100M application.

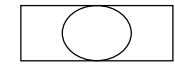




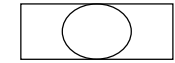
Touchpad control



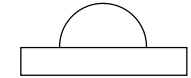
LED HSMD-C170 ORANGE



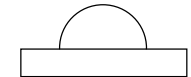
LED HSMB-C172 BLUE



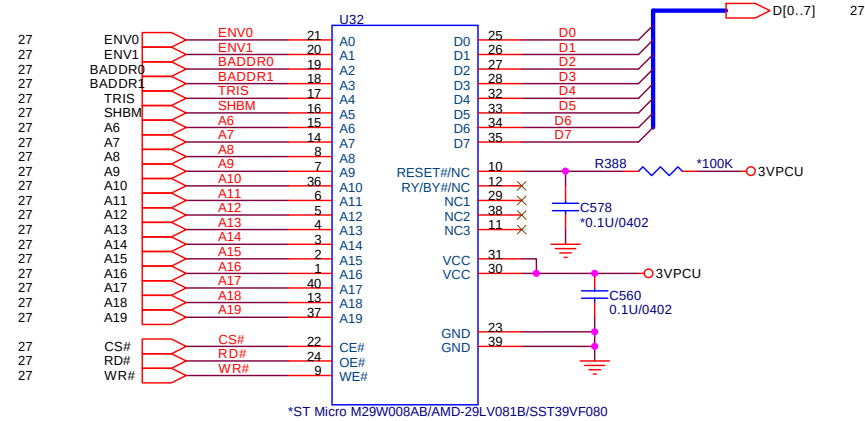
LED HSMD-C110 ORANGE



LED HSMD-C112 BLUE



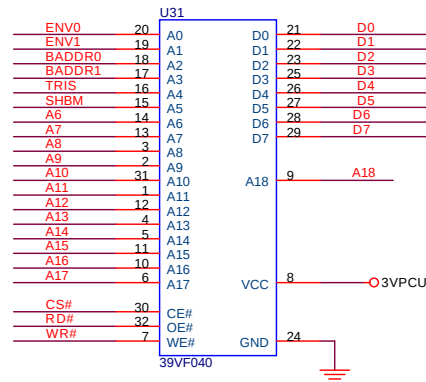
8Mbit (1M Byte), TSSOP40



- 1.AMD-29LV081B require MAX 500nS Tready for it's hardware reset.And MAX6326_UR29 has >100mS reset timing.So we can tie it's reset# pin to +3VALW directly.
- 2.SIO has internal 20 mS delay of VCC1_PWROK

AMD :Pin 10 is RESET# ; Pin12 is RY/BY#
SST :Pin10,12 are NC

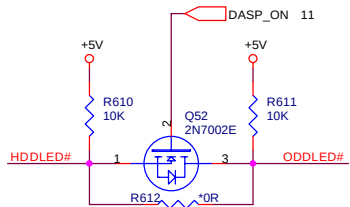
4Mbit (512k Byte), TSSOP32



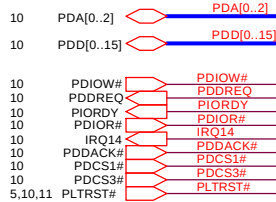
SI stage:
Add PLCC32 cause of it is convenient for Bios debugging.
Sting 10/01/2004

PV-II stage:
Delete PLCC32 for SMT request
Mika 02/03/2005

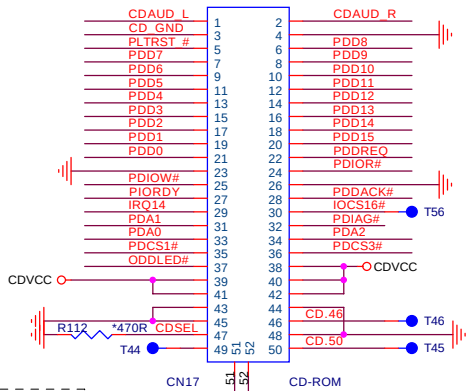
HDD, CD-ROM



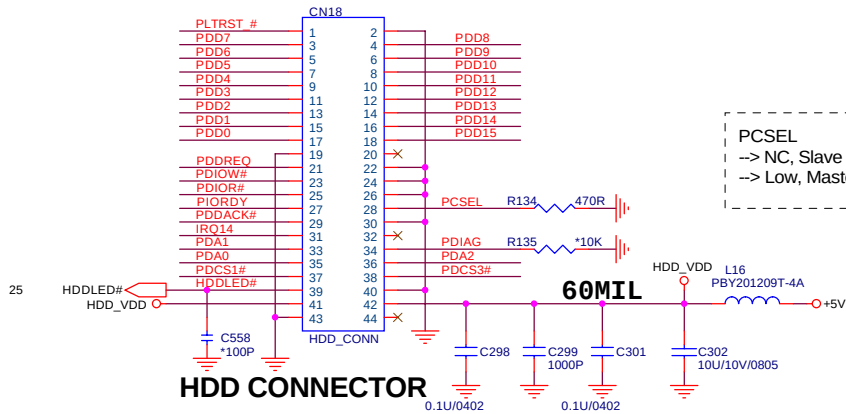
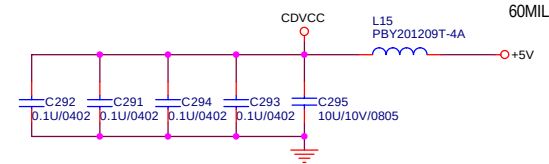
ADD DASP_ON FOR IDE CABLE SELECT



CD-ROM

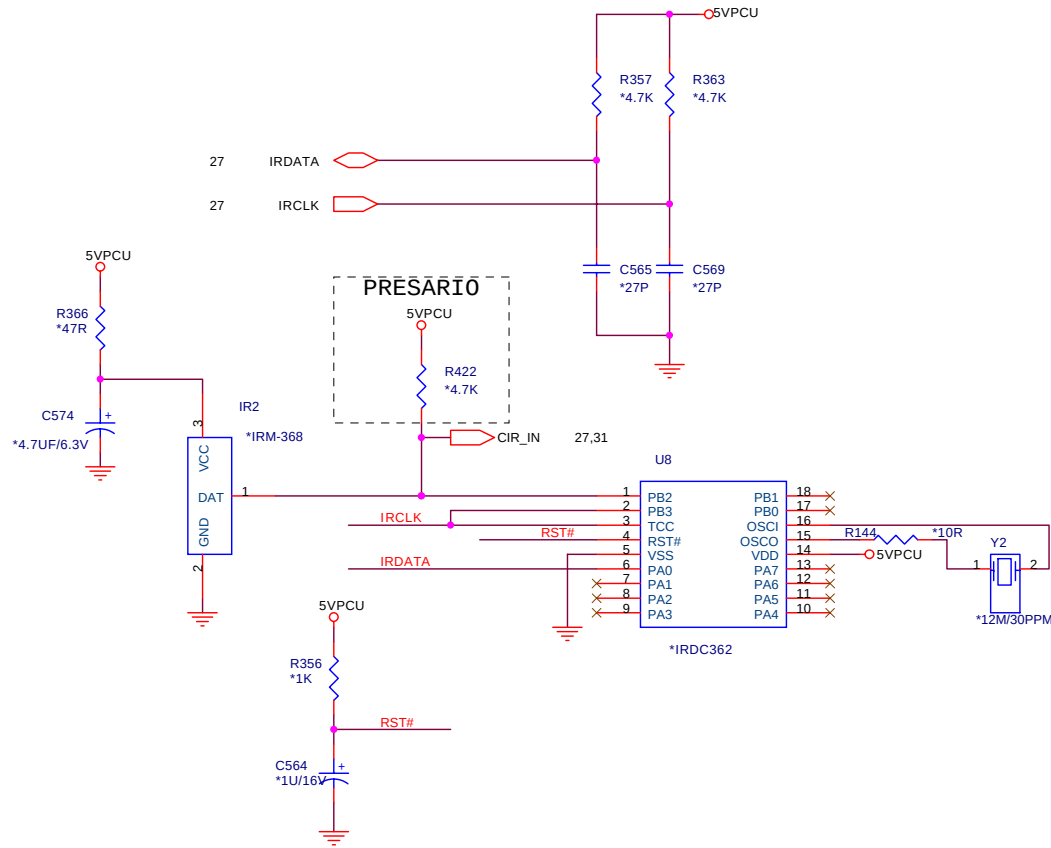


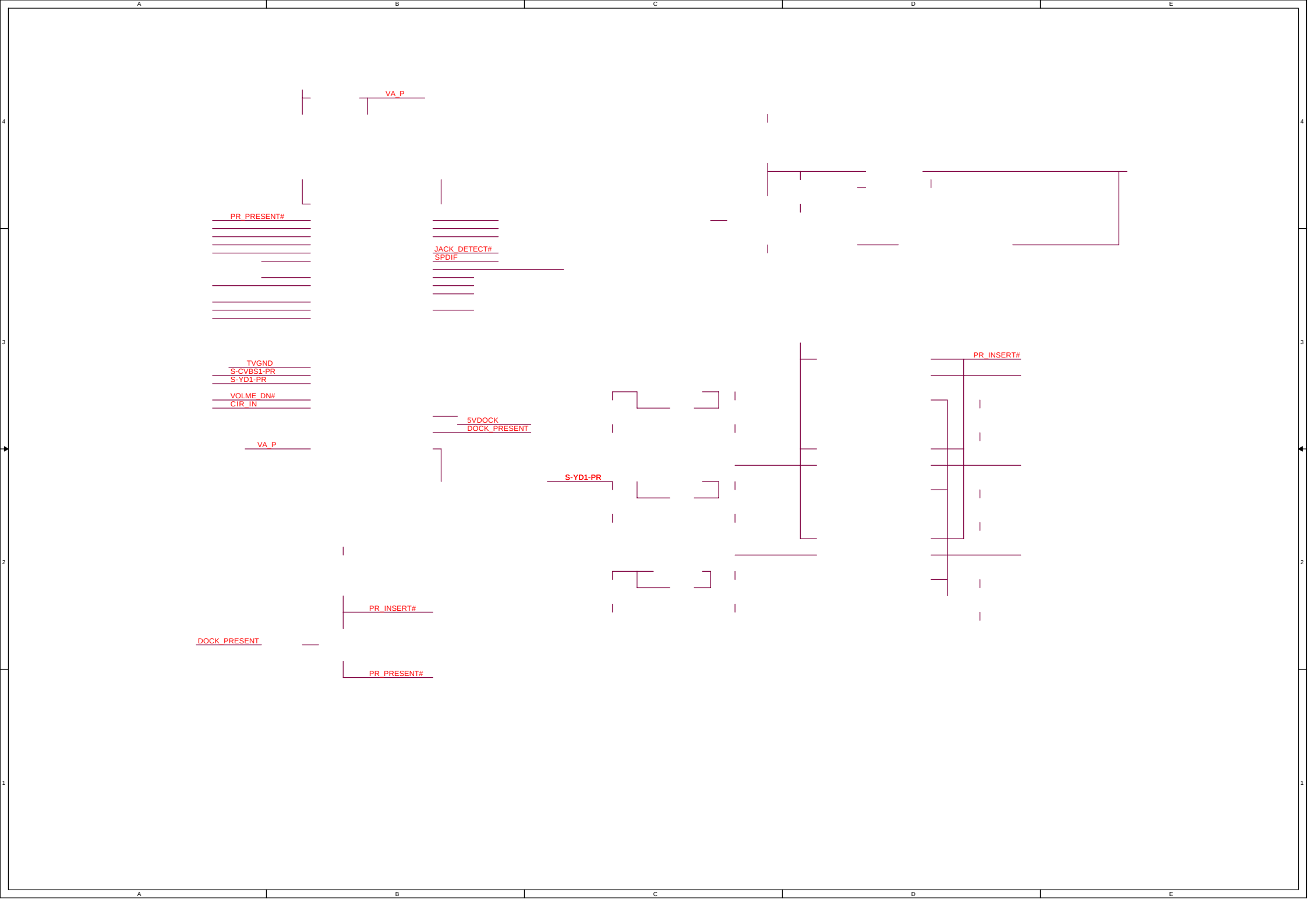
CDSSEL
--> NC, Slave device
--> Low, Master device



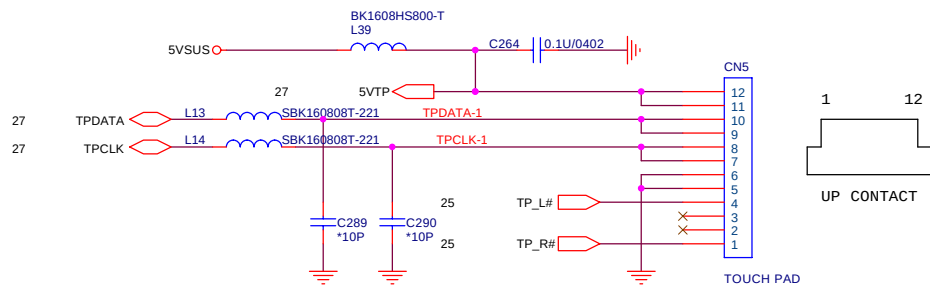
PCSEL
--> NC, Slave device
--> Low, Master device

HDD CONNECTOR

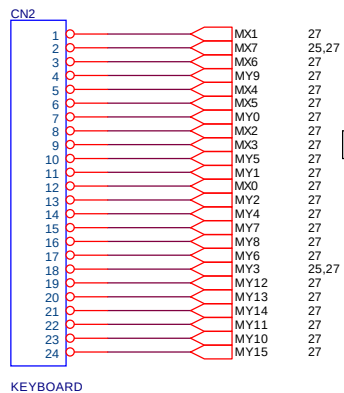




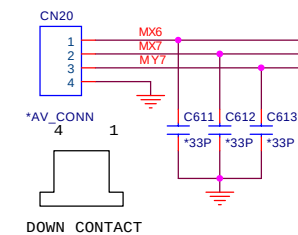
TOUCH PAD CONNECTOR



KEYBOARD CONNECTOR

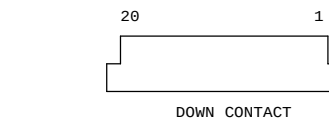
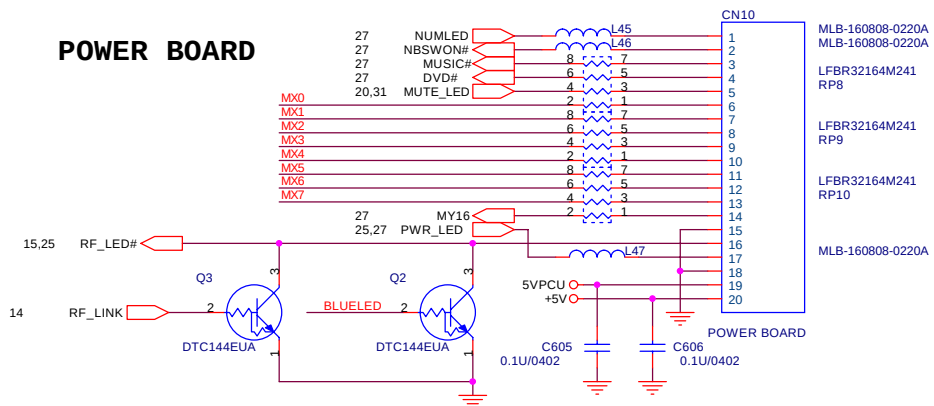


AV BOARD



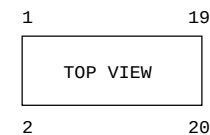
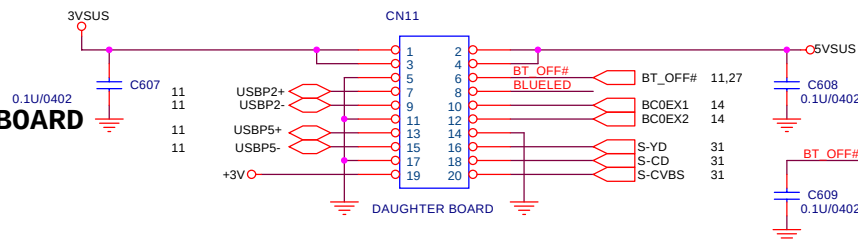
MX6	MX7
ENTER	MENU

POWER BOARD



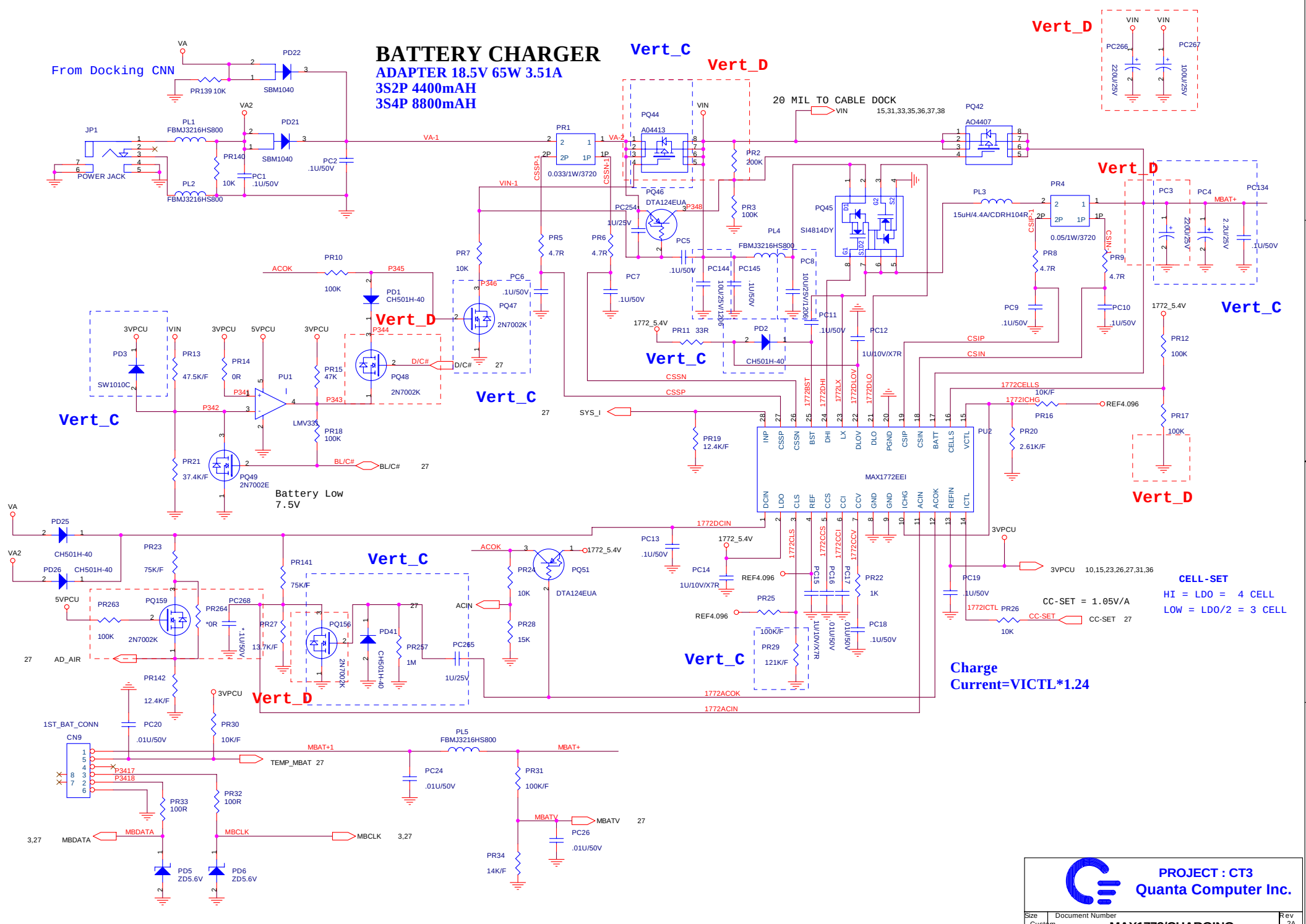
MX0	MX1	MX2	MX3	MX4	MX5	MX6	MX7
BACK	PLAY/PAUSE	FORWARE	STOP	VOL UP	MUTE	VOL DN	WIRELESS

DAUGHTER BOARD



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Quanta Computer Inc.

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BATTERY CHARGER
ADAPTER 18.5V 65W 3.51A
3S2P 4400mAh
3S4P 8800mAh

Vert_D

Vert_C

Vert_D

Vert_D

Vert_C

Vert_C

Vert_C

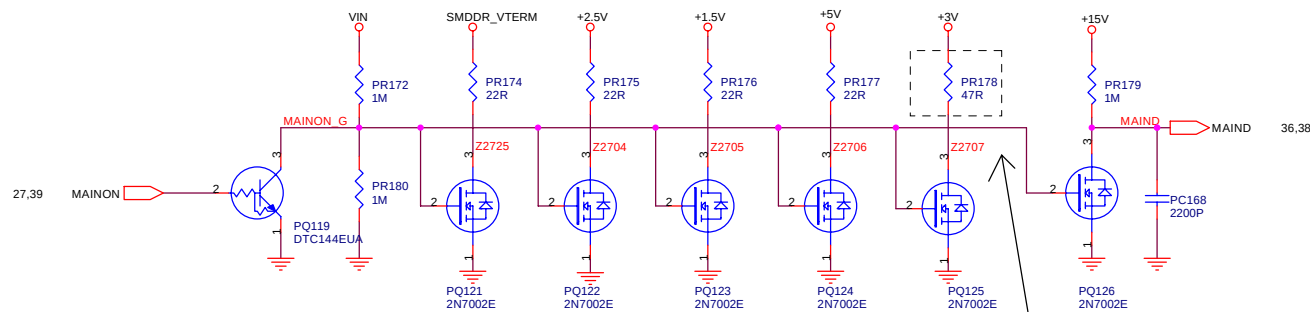
Vert_C

Vert_C

Vert_D

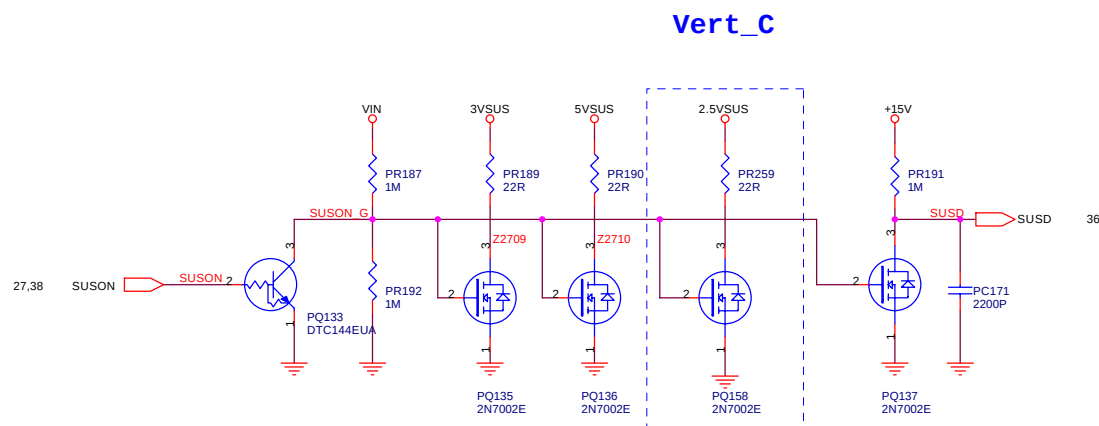
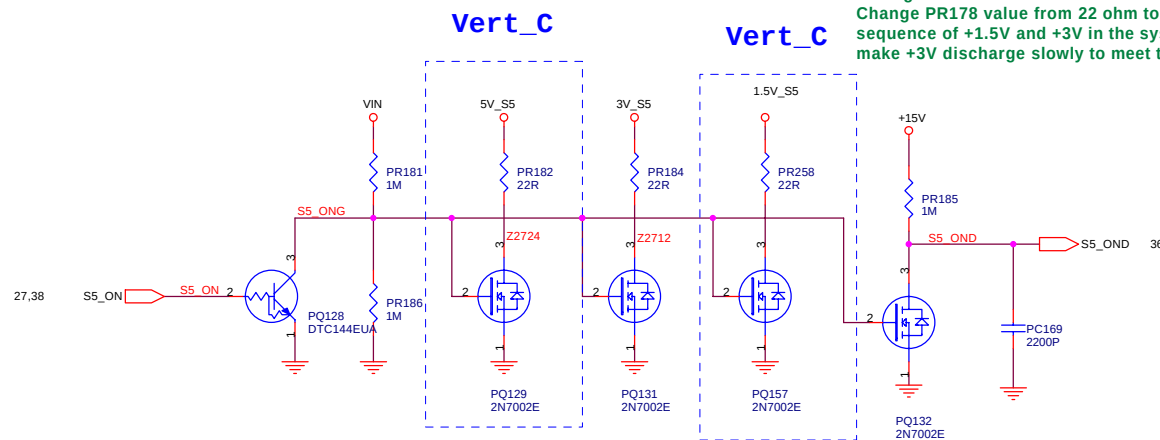
Charge
Current=VICTL*1.24

CELL-SET
HI = LDO = 4 CELL
LOW = LDO/2 = 3 CELL



SI stage:
Change PR178 value from 22 ohm to 47 ohm, Intel has required the timing sequence of +1.5V and +3V in the system, we'd like to increase the value to make +3V discharge slowly to meet the specification of falling time.

Sting 10/19/2004



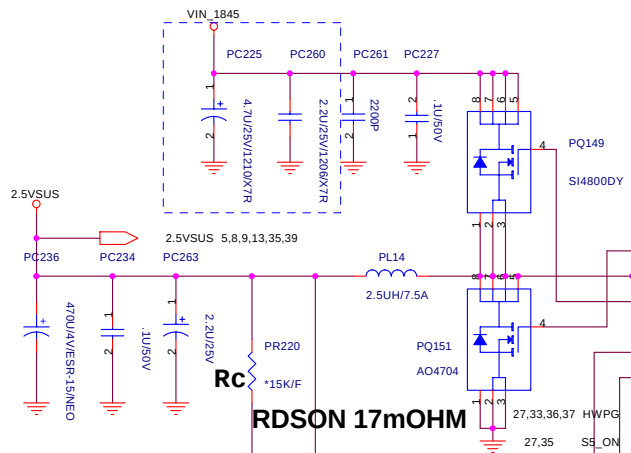
PROJECT : CT3
Quanta Computer Inc.

Size	Document Number	Rev
Custom		2A
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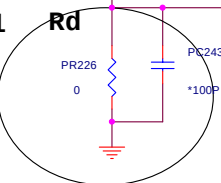
DISCHARGE

2.5VSUS
6.5A
S0-S3

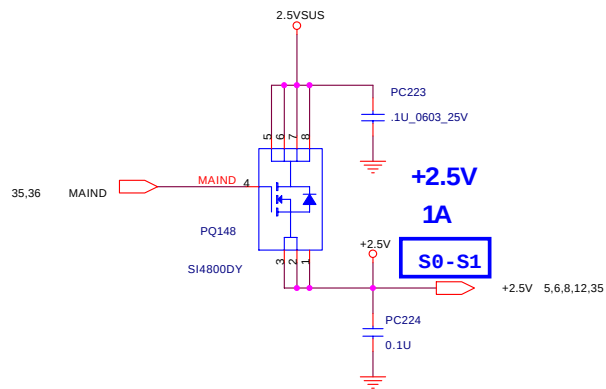
Vert_C



$$V_{out} = (1 + R_c/R_d) * 1$$

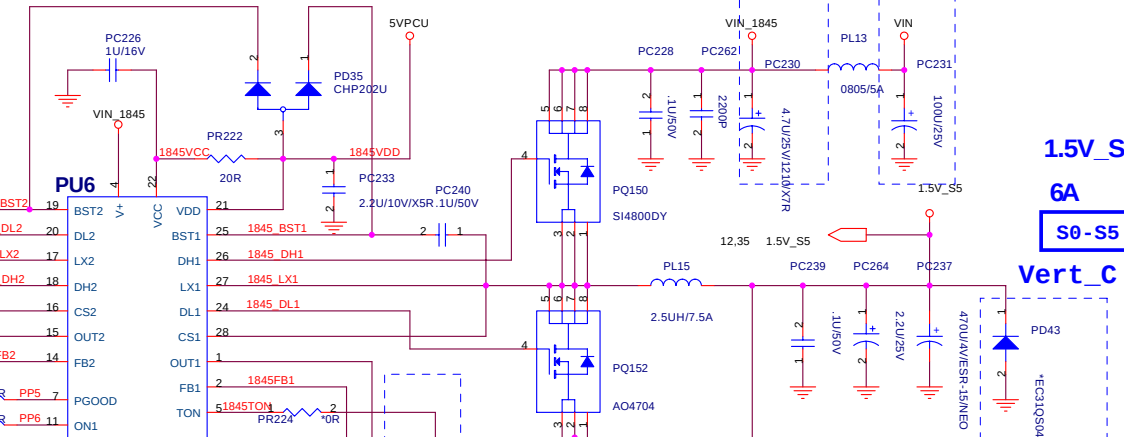


Fix 2.5V Output



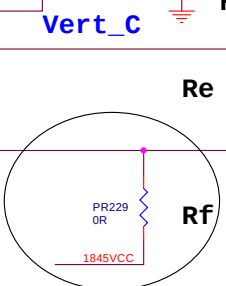
+2.5V
1A
S0-S1

Vert_C Vert_C

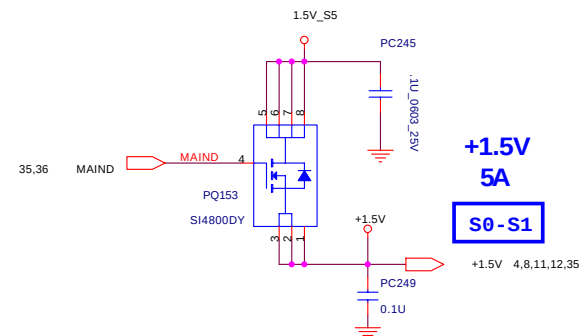


1.5V_S5
6A
S0-S5

Vert_C



Fix 1.5V Output



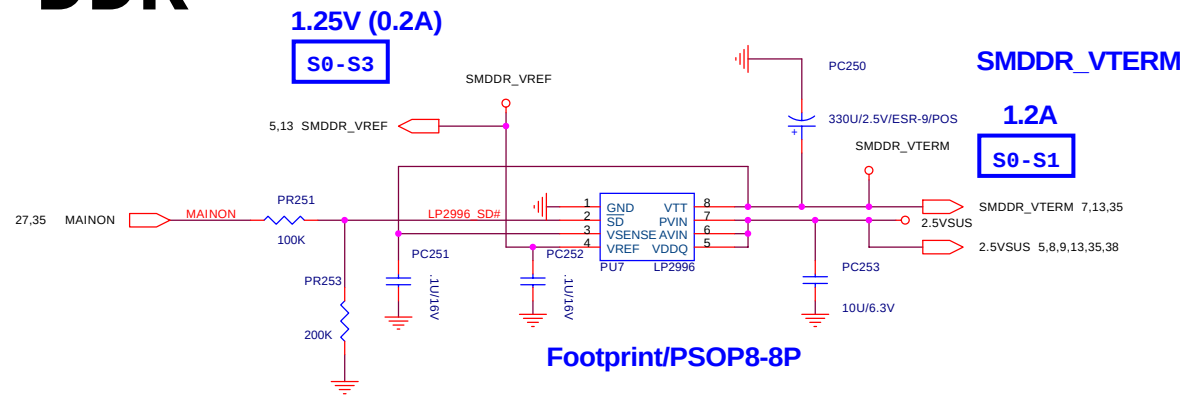
+1.5V
5A
S0-S1



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DDR



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MODEL	REV	CHANGE LIST	Model	CT3/5 MB BOARD	
			Page	FROM	TO
CT3/5 MB 31CT3MB0015 31CT3MB0031	1A	PAGE 2 --- Enable CLK48M from clock generator for the PLL circuit of 7411, and disable the ocsillator circuit of PCI7411 PLL. PAGE 3 --- Remove H/W shutdown circuit that supported ADM1032. PAGE 4 --- Use X7R type to replace Y5V type for CPU Decoupling/Bypass capacitor. PAGE 10 --- Add a terminal resistor R706 for -CODE_RST# to improve signal quality. PAGE 11 --- 1. Add a 10K pull-up resistor on MCH_SYNC# for booting. 2. Change the power plane of PCIE_WAKE# from 3VSUS to 3V_S5 to solve system can't turn off issue. 3. Change the power plane of ICH_THRM# and SCI_# from 3VSUS to +3V to reduce leakage. PAGE 12 --- We can also use 5VSUS to instead of 5V_S5 to save cost of MOSFET(A06402). PAGE 15 --- Add a level-shift cicuit for EDID interface. PAGE 17 --- 1. Add a off-page and a EMI solution for CLK48M. 2. Remove the reserve resistors (R693-R695) of parallel interface for PCI1510. PAGE 18 --- Remove R696, connect controller and power switch directly . PAGE 19 --- Change R682&R683 value from 56 ohms to 0 ohm cause of BOM error at A-test. PAGE 22 --- 1. Change MC3 type from Y5V to X7R to improve singal quality. 2. Connect H1/H3 to AGND via a 0 ohm resistor by Conexant's comment. PAGE 23 --- 1. Add a terminal resistor R707 for RTL8100/8110 id selection. 2. Add a 0.1uF to make Q40 turn on slowly to aviod 3VPCU drop issue. PAGE 24 --- Modified transformer circuit cause of CT can't connect each other on 10/100M application. PAGE 26 --- Add a flashrom as PLCC32 type for BIOS debugging. PAGE 27 --- 1. Change R352 value from 120K ohms to 20M ohms. 2. Add a LPC debug port for software team to debug convenient. PAGE 30 --- Add GMT fan controller for B-test to costdown. PAGE 31 --- 1. Add ESD protection circuit for S-VIDEO signal to Docking. 2. Add R713 to enable the mux in the Tampa-2 cable PAGE 33 --- Change PR143 value from 100K to 10K to solve display abnormal issue. PAGE 35 --- 1. Move 5V_S5 circuit to Page 36. 2. De-popuplate PQ129 and PR182. 3. Change PR178 value from 22 ohm to 47 ohm. PAGE 36 --- Remove PC170 and PQ127 but reserve 5V_S5 power circuit.	1	1A	
			2	1A	2A
			3	1A	2A
			4	1A	2A
			5	1A	
			6	1A	
			7	1A	
			8	1A	
			9	1A	
			10	1A	2A
			11	1A	2A
			12	1A	2A
			13	1A	
			14	1A	
			15	1A	2A
			16	1A	
			17	1A	2A
			18	1A	2A
			19	1A	2A
			20	1A	
			21	1A	
			22	1A	2A
			23	1A	2A
			24	1A	2A
			25	1A	
			26	1A	2A
			27	1A	2A
			28	1A	
			29	1A	
			30	1A	2A
			31	1A	2A
			32	1A	
			33	1A	2A
			34	1A	
			35	1A	2A
			36	1A	2A
			37	1A	
			38	1A	
			39	1A	
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MODEL	REV	CHANGE LIST	Model	CT3/5 MB BOARD	
			Page	FROM	TO
CT3/5 MB 31CT3MB0015 31CT3MB0031	2A	PAGE 2 --- 1. Add C1048 for CLK48M to get better EMI performance. PAGE 3 --- 1. Add R733 as pull-up resistor for PREQ#. PAGE 11 --- 1. Add RF_OFF# and BT_OFF# circuit. PAGE 17 --- 1. Populate R704 and C1046 to get better EMI performance. 2. Remove R701 & R702 for unused PCI1510RVGF circuit. PAGE 18 --- 1. Disconnect SM_PHYS_WP on controller side. 2. Tie SM_EL_WP with SM_PHYS_WP on conn side to allow for normal operation of SD and SM. 3. Add a discharge circuit for media card power. 4. Add R718 to solve cross-talk issue of MS-Pro card. 5. Add R717 to solve SM card can't write protect issue. 6. Add R719~R736 as terminal on all multi-function pins. 7. Add pull-up circuit. PAGE 27 --- 1. Reserve 0R for RF_OFF# and BT_OFF# circuit. 2. Modify LPC pin name. PAGE 28 --- 1. Change HDD and ODD select definition. PAGE 30 --- Adjust Capacitors and Bead to improve CRT timing issue. 1. Change L66, L67, L68 from BK1608HM470 to 0R. 2. Remove C931, C932, C933. 3. Change C934, C935, C936 from 22P to 5.6P. 4. Change C6, C14, C350 from 10P to 5.6P. 5. Change L1, L26, L27 from BK1608HM470 to BLM18BA750SN1T. PAGE 31 --- 1. Change L5,L6,L7,C57,C58,C64,C77,C113,C121 value to improve S-video quality. 2. Reserve S-video impedance match circuit.	1	1A	
			2	2A	3A
			3	2A	3A
			4	2A	
			5	1A	
			6	1A	
			7	1A	
			8	1A	
			9	1A	
			10	2A	
			11	2A	
			12	2A	
			13	1A	
			14	1A	
			15	2A	
			16	1A	
			17	2A	3A
			18	2A	3A
			19	2A	
			20	1A	
			21	1A	
			22	2A	
			23	2A	
			24	2A	
			25	1A	
			26	2A	2A
			27	2A	
			28	1A	2A
			29	1A	
			30	2A	
			31	2A	
			32	1A	
			33	2A	
			34	2A	
			35	2A	
			36	2A	
			37	2A	
			38	2A	
			39	2A	
			 PROJECT : CT3 Quanta Computer Inc. Size Custom Document Number Date: Friday, March 04, 2005 SI history Sheet 41 of 42 Rev 1A		

MODEL	REV	CHANGE LIST	Model	CT3/5 MB BOARD	
			Page	FROM	TO
CT3/5 MB 31CT3MB0015 31CT3MB0031	3A	PAGE 08 --- 1.To add a LDO for clear POWER to solve TV problem PAGE 10 --- 1.EMI team suggest us to add a Bead and a Capacity to improve EMI PAGE 14 --- 1.TO change net name from BITCLK to BITCLK_SB. 2.To add a 10U Capacity for better power. PAGE 18 --- 1.To add a circuit for MS PRO DUO problem. 2.To add a discharge circuit for media card power. 3.To change resistor value to solve card-reader problem. PAGE 19 --- 1.To delete CML1, CML2, L1 and L37 for SMT request. 2.To change USB and 1394 connector for strong ME. PAGE 26 --- 1.To delete PLCC32 for SMT request	1	1A	
			2	3A	
			3	3A	
			4	2A	
			5	1A	
			6	1A	
			7	1A	
			8	1A	2A
			9	1A	
			10	2A	3A
			11	2A	
			12	2A	
			13	1A	
			14	1A	2A
			15	2A	
			16	1A	
			17	3A	
			18	3A	4A
			19	2A	3A
			20	1A	
			21	1A	
			22	2A	
			23	2A	
			24	2A	
			25	1A	
			26	2A	3A
			27	2A	
			28	2A	
			29	1A	
			30	2A	
			31	2A	
			32	1A	
			33	2A	
			34	2A	
			35	2A	
			36	2A	
			37	2A	
			38	2A	
			39	2A	
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